F021 Flash Module Controller (FMC)

The Flash electrically-erasable programmable read-only memory module is a type of nonvolatile memory that has fast read access times and is able to be reprogrammed in the field or in the application. This chapter describes the F021 Flash module controller (FMC).

Topic **Page**

5.1 Overview

The F021 Flash is used to provide non-volatile memory for instruction execution or data storage. The Flash can be electrically programmed and erased many times to ease code development.

Refer to the following documents for support in how to initialize and use the on-chip Flash and its API:

- *Initialization of Hercules ARM Cortex-R4F Microcontrollers Application Report* (SPNA106)
- *F021 (Texas Instruments 65nm Flash) Flash API Reference Guide* (SPNU501)

5.1.1 Features

- Read, program and erase with a single 3.3 V supply voltage
- Supports error detection and correction
  - Single Error Correction and Double Error Detection (SECDED)
  - Error Correction Code (ECC) is evaluated in the CPU for the main Flash bank arrays and in the Flash Wrapper for the EEPROM emulation Flash banks
  - Address bits included in ECC calculation
- Provides different read modes to optimize performance and verify the integrity of Flash contents
- Provides built-in power mode control logic
- Integrated program/erase state machine
  - Simplifies software algorithms
  - Supports simultaneous read access on a bank while performing a write or erase operation on any one of the remaining banks
  - Suspend command allows read access to a sector being programmed/erased
  - Fast erase and program times (for details, see the device-specific data sheet)

For the actual size of the Flash memory for the device, see the device-specific data sheet.

5.1.2 Definition of Terms

Terms used in this document have the following meaning:

- ATCM: Port A tightly coupled memory
- BAGP (Bank Active Grace Period): Time (in HCLK cycles) from the most recent Flash access of a particular bank until that bank enters fallback power mode. This reduces power consumption by the Flash. However, it can also increase access time.
- bw: Normal data space bank data width of a Flash bank. The bw is 128 bits (144 bits including the error correction bits).
- bwe: EEPROM emulation bank is 128-bits wide (144 bits including the error correction bits).
- Charge pump: Voltage generators and associated control (logic, oscillator, and bandgap, for example).
- CSM: Program/erase command state machine
- Fallback power mode: The power mode (active, standby or sleep, depending on which mode is selected) into which a bank or the charge pump falls back each time the active grace period expires.
- Flash bank: A group of Flash sectors that share input/output buffers, data paths, sense amplifiers, and control logic.
- FEE - Flash EEPROM Emulation. Features on the FMC to support using a Flash type memory in place of an EEPROM Flash memory. EEPROM is erasable by the word while this Flash memory is only erasable by the sector. The FEE bank is accessible only through Bus 2 in a special address range and always resides in bank 7.
- Flash module: Flash banks, charge pump, and Flash wrapper.
- Flash wrapper: Power and mode control logic, data path, wait logic, and write/erase state machines.
- FMC: Flash Module Controller.
- Command: A sequence of coded instructions to Flash module to execute a certain task.

- FSM (Flash State Machine): State machine that parses and decodes FSM commands. It executes embedded algorithms and generates control signals to both Flash bank and charge pump during the actual program/erase operation.
- OTP (one-time programmable): A program-only-once Flash sector (cannot be erased)
- PAGP (Pump Active Grace Period): Time (in HCLK cycles) from when the last of the banks have entered fallback power mode until the pump enters a fallback power mode. This can reduce power consumption by the Flash; however, it can also increase access time.
- Pipeline mode: The mode in which Flash is read 128 bits (+ 16 bit ECC) at a time, providing higher throughput.
- Sector: A contiguous region of Flash memory which must be erased simultaneously.
- Wide_Word: The width of the data output from the Flash bank. This is 144-bits wide for main Flash and for the FEE bank.
- Standard read mode: The mode assumed when the pipeline mode is disabled. Physically, 128 (+ 16 bit ECC) is read at a time. However, only 32 bits of data is used while the other bits of data are discarded.
- Read Margin 1 mode: More stringent read mode designed for early detection of marginally erased bits.
- Read Margin 0 mode: More stringent read mode designed for early detection of marginally programmed bits.

5.1.3 F021 Flash Tools

Texas Instruments provides the following tools for F021 Flash:

- nowECC Generation Tool - to generate the Flash ECC from the Flash data.
- nowFLASH Programming Tool - to erase/program/verify the device Flash content through JTAG.
- Code Composer Studio - the development environment with integrated Flash programming capabilities.
- F021 Flash API Library - a set of software peripheral functions to program/erase the Flash module. Refer to *F021 Flash API Reference Guide* (SPNU501) for more information.

5.2 Default Flash Configuration

At power up, the Flash module state exhibits the following properties:

- Wait states are set to 1 data wait state and 0 address wait states
- Pipeline mode is disabled
- The Flash content is protected from modification
- Power modes are set to *Active* (no power savings)
- The boot code must initialize the wait states (including data wait states and address wait states) and the desired pipeline mode by initializing the FRDCNTL register to achieve the optimum system performance. This needs to be done before switching to the final device operating frequency. Refer to *Initialization of Hercules ARM Cortex-R4F Microcontrollers Application Report* (SPNA106) for more information.

5.3 SECDED

The Flash memory can be protected by Single Error Correction Double Error Detection (SECDED). The main program memory is protected by the SECDED circuit inside of the Cortex-R4 CPU. All OTP and the FEE memory (bank 7) is protected by SECDED logic in the Flash wrapper.

5.3.1 SECDED Initialization

Flash error detection and correction is not enabled at reset. To enable SECDED, error correction detection must be enabled in the Flash wrapper, the CPU event bus must be enabled and SECDED must be enabled within the CPU. Refer to *Initialization of Hercules ARM Cortex-R4F Microcontrollers Application Report* (SPNA106) for information on these steps.

The ECC values for all of the ATCM program memory space (Flash banks 0 through 6) must be programmed into the Flash before SECDED is enabled. This can be done by generating the correct values of the ECC with an external tool such as nowECC or may be generated by the programming tool. The Cortex-R4 CPU may generate speculative fetches to any location within the ATCM memory space. A speculative fetch to a location with invalid ECC, which is subsequently not used, will not create an abort, but will set the ESM flags for a correctable or uncorrectable error. An uncorrectable error will unconditionally cause the nERROR pin to toggle low. Therefore care must be taken to generate the correct ECC for the entire ATCM space including the holes between sections and any unused or blank Flash areas.

The Cortex-R4 CPU does not generate speculative fetches into the address space of bank 7, the EEPROM Emulation Flash. It is only necessary to initialize the ECC values of the locations which will be intentionally read by the CPU or other bus masters.

5.3.2 ECC Encoding

Nineteen address lines are also included in the ECC calculation. A failure of a single address line inside of the bank will be treated as an uncorrectable error. The ECC encoding is shown in Table 5-1. Bits 31:0 come from the word at the address ending in 0x0 or 0x8, Bits 63:31 come from the word at the address ending in 0x4 or 0xC.

Table 5-1. ECC Encoding for LE Devices

		82	81	80	79	78	77	76	75	74	73	72	71	70	69	68	67	66	65	64
		Participating Address Bits																		
ADDR_MSW_LSW	**ECC Bit**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**09**	**08**	**07**	**06**	**05**	**04**	**03**
0007F_00FFFF00_FF0000FF	7													x	x	x	x	x	x	x
7FF80_FF0000FF_FF0000FF	6	x	x	x	x	x	x	x	x	x	x	x	x							
07F80_FF00FF00_FF00FF00	5					x	x	x	x	x	x	x	x							
19F83_C0FCC0FC_C0FCC0FC	4			x	x			x	x	x	x	x	x						x	x
6A78D_38E338E3_38E338E3	3	x	x		x		x			x	x	x	x				x	x		x
2A9B5_A699A699_A699A699	2		x		x		x		x			x	x		x	x		x		x
0BAD1_15571557_15571557	1				x		x	x	x		x		x	x		x				x
554EA_B4D1B4D1_4B2E4B2E	0	x		x		x		x		x			x	x	x		x		x	

Participating Data Bits																																				
63	**62**	**61**	**60**	**59**	**58**	**57**	**56**	**55**	**54**	**53**	**52**	**51**	**50**	**49**	**48**	**47**	**46**	**45**	**44**	**43**	**42**	**41**	**40**	**39**	**38**	**37**	**36**	**35**	**34**	**33**	**32**	**31**	**30**	**29**	**28**	**27**
								x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x									x	x	x	x	x
x	x	x	x	x	x	x	x																	x	x	x	x	x	x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x									x	x	x	x	x	x	x	x									x	x	x	x	x
x	x							x	x	x	x	x	x			x	x							x	x	x	x	x	x			x	x			
		x	x	x				x	x	x				x	x			x	x	x				x	x	x				x	x			x	x	x
x		x			x	x		x			x	x			x	x		x			x	x		x			x	x			x	x		x		
			x		x		x		x		x		x	x	x				x		x		x		x		x		x	x	x				x	
x		x	x		x			x	x		x				x	x		x	x		x			x	x		x				x		x			x

Participating Data Bits																											Parity (1)	Check Bits (2)
26	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**09**	**08**	**07**	**06**	**05**	**04**	**03**	**02**	**01**	**00**		
x	x	x																	x	x	x	x	x	x	x	x	Even	ECC[7]
x	x	x																	x	x	x	x	x	x	x	x	Even	ECC[6]
x	x	x									x	x	x	x	x	x	x	x									Even	ECC[5]
			x	x	x	x	x	x			x	x							x	x	x	x	x	x			Even	ECC[4]
			x	x	x				x	x			x	x	x				x	x	x				x	x	Odd	ECC[3]
x	x		x			x	x			x	x		x			x	x		x			x	x			x	Odd	ECC[2]
x		x		x		x		x	x	x				x		x		x		x		x		x	x	x	Even	ECC[1]
	x	x			x		x	x	x			x			x		x	x			x		x	x	x		Even	ECC[0]

(1) For Odd parity, XOR a 1 to the row's XOR result. For even Parity, use the row's XOR result directly.
(2) Each ECC[x] bit represents the XOR of all the address and data bits marked with x in the same row.

5.3.3 Syndrome Table: Decode to Bit in Error

The syndrome is an 8-bit value that decodes to the bit in error. The bit in error can be a bit among the 64 data bits, the 19 address bits, or a bit among the 8 ECC check bits. A syndrome value of 00000000 indicates there is no error. Any other syndrome combinations not shown in the table are uncorrectable multi-bit error. Errors of three of more bits may escape detection. The syndrome decoding is shown in Table 5-2.

Table 5-2. Syndrome Table, Decode to Bit in Error

Address Bit Error Position																		
21	20	19	18	17	16	15	14	13	12	11	10	09	08	07	06	05	04	03
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0
0	0	1	1	0	0	1	1	1	1	1	1	0	0	0	0	0	1	1
1	1	0	1	0	1	0	0	1	1	1	1	0	0	0	1	1	0	1
0	1	0	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0	1
0	0	0	1	0	1	1	1	0	1	0	1	1	0	1	0	0	0	1
1	0	1	0	1	0	1	0	1	0	0	1	1	1	0	1	0	1	0

Data Bit Error Position																																				
63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	31	30	29	28	27
0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1
1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1
1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	0	0	0
0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	1	1	1
1	0	1	0	0	1	1	0	1	0	0	1	1	0	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	0	1	1	0	1	0	0
0	0	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	0	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	0	0	1	0
1	0	1	1	0	1	0	0	1	1	0	1	0	0	0	1	1	0	1	1	0	1	0	0	1	1	0	1	0	0	0	1	0	1	0	0	1

Data Bit Error Position																											ECC Error Bit								
26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	09	08	07	06	05	04	03	02	01	00	07	06	05	04	03	02	01	00	
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	Bit[7]
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	Bit[6]
1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	Bit[5]
0	0	0	1	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	Bit[4]
0	0	0	1	1	1	0	0	0	1	1	0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	0	0	1	0	0	0	Bit[3]
1	1	0	1	0	0	1	1	0	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	0	1	0	0	0	0	0	1	0	0	Bit[2]
1	0	1	0	1	0	1	0	1	1	1	0	0	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	0	0	0	0	0	1	0	Bit1]
0	1	1	0	0	1	0	1	1	1	0	0	1	0	0	1	0	1	1	0	0	1	0	1	1	1	0	0	0	0	0	0	0	0	1	Bit[0]

5.3.4 Syndrome Table: An Alternate Method

Table 5-3. Alternate Syndrome Table

Syndrome lsb: 3:0	Syndrome msb 7:4															
	0x	**1x**	**2x**	**3x**	**4x**	**5x**	**6x**	**7x**	**8x**	**9x**	**Ax**	**Bx**	**Cx**	**Dx**	**Ex**	**Fx**
x0	good	E04	E05	D	E06	D	D	D62	E07	D	D	D46	D	M	M	D
x1	E00	D	D	D14	D	A19	A17	D	D	A04	M	D	M	D	D	D30
x2	E01	D	D	M	D	D34	D56	D	D	D50	D40	D	M	D	D	M
x3	D	D18	D08	D	M	D	D	A15	A09	D	D	M	D	D02	D24	D
x4	E02	D	D	D15	D	D35	D57	D	D	D51	D41	D	M	D	D	D31
x5	D	D19	D09	D	M	D	D	D63	A08	D	D	D47	D	D03	D25	D
x6	D	D20	D10	D	M	D	D	A14	A07	D	D	M	D	D04	D26	D
x7	M	D	D	M	D	D36	D58	D	D	D52	D42	D	M	D	D	M
x8	E03	D	D	M	D	D37	D59	D	D	D53	D43	D	M	D	D	M
x9	D	D21	D11	D	A21	D	D	A13	A06	D	D	M	D	D05	D27	D
xA	D	D22	D12	D	D33	D	D	A12	D49	D	D	M	D	D06	D28	D
xB	D17	D	D	M	D	D38	D60	D	D	D54	D44	D	D01	D	D	M
xC	D	D23	D13	D	A20	D	D	A11	A05	D	D	M	D	D07	D29	D
xD	M	D	D	M	D	D39	D61	D	D	D55	D45	D	M	D	D	M
xE	D16	D	D	M	D	A18	A16	D	D	A03	M	D	D00	D	D	M
xF	D	M	M	D	D32	D	D	A10	D48	D	D	M	D	M	M	D

- E0x - Single-bit ECC error, correctable
- Dxx - Single-bit data error, correctable
- Axx - Single-bit address error, uncorrectable
- D - Double-bit error, uncorrectable
- M - Multi-bit errors, uncorrectable

5.4 Memory Map

The Flash module contains the program memory, which is mapped starting at location 0, and one Customer OTP sector and one TI OTP sector per bank. The Customer OTP sectors may be programmed by the customer, but cannot be erased. They are typically blank in new parts. The TI OTP sectors are used to contain manufacturing information. They may be read by the customer but can not be programmed or erased. The TI OTP sectors contain settings used by the Flash API to setup the Flash state machine for erase and program operations.

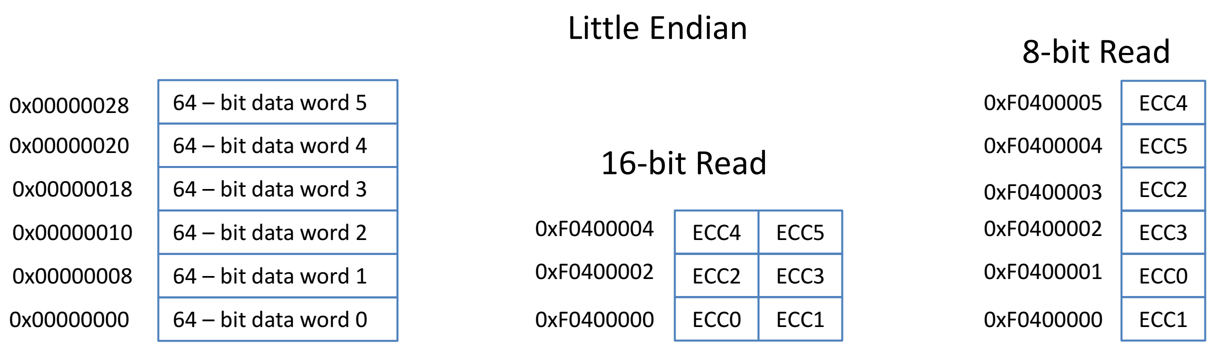
All of these OTP regions are memory-mapped to facilitate ease of access by the CPU. They are memory mapped to an offset starting at F000 0000h in the CPU's memory map.

The RWAIT value is used to define the number of wait states for the program-memory Flash. The EWAIT value is used to define the number of wait states for the data Flash in bank 7. Bank 7 starting at offset F020 0000h is dedicated for data storages such as EEPROM Emulation.

5.4.1 Location of Flash ECC Bits

The Flash ECC bits can be read starting at address 0xF0400000. The ECC bits are packed in their memory space as shown in Figure 5-1. The ECC bytes must be read as bytes or halfwords. Reading a single ECC byte with ECC enabled will actually cause 144 bits to be read from the Flash, and the ECC bits will be corrected if necessary. Any errors in either of the two double-words accessed will be recorded in the FEDACSTATUS register (main memory) or the EE_STATUS register (bank 7).

Figure 5-1. ECC Organization for Program Flash (144-Bits Wide)



5.4.2 OTP Memory

5.4.2.1 Flash Bank and Sector Sizes

Flash Bank/Sectoring information can be determined from the device-specific datasheet or can be computed by reading locations in the TI OTP and FMC registers.

The number of banks, which banks are available, and the number of sectors for bank 0 can be read from TI OTP location F008 0158h as shown in Figure 5-2 and described in Table 5-4.

Figure 5-2. TI OTP Bank 0 Sector Information

31-24	23-16
Reserved	BX_NUM_Sectors
R	R

15	14	13	12	11	10	9	8	7-0
B7	B6	B5	B4	B3	B2	B1	B0	NUM_Banks
R-1	R-0	R-0	R-0	R-0	R-0	R-1	R-1	R

LEGEND: R = Read only

Table 5-4. TI OTP Bank 0 Sector Information Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reserved. All bits will be read as 0.
23-16	BX_NUM_Sectors	1-32	Number of sectors in this bank.
15	B7	1	1 = Bank 7 is present
14	B6	0	0 = Bank 6 is not present
13	B5	0	0 = Bank 5 is not present
12	B4	0	0 = Bank 4 is not present
11	B3	0	0 = Bank 3 is not present
10	B2	0	0 = Bank 2 is not present
9	B1	1	1 = Bank 1 is present
8	B0	1	1 = Bank 0 is present
7-0	NUM_Banks	2 or 3	Number of banks on this part.

The bank sector information is repeated once for each bank in the device. The number of sectors is unique for each bank. The number of banks and which banks are implemented is repeated in each location. Use the TI OTP information for bank 0 to determine which banks are in the device, and then read the number of sectors for each bank using the TI OTP locations shown in Table 5-5.

Table 5-5. TI OTP Sector Information Address

Bank	TI OTP Address
0	F008 0158h
1	F008 2158h
2	F008 4158h
3	F008 6158h
4	F008 8158h
5	F008 A158h
6	F008 C158h
7	F008 E158h

5.4.2.2 Package and Memory Size

Package and memory size information can be determined from the device-specific datasheet, or can be computed by reading locations in the TI OTP Bank 0 registers.

The package and memory size can be read from TI OTP location F008 015Ch as shown in Figure 5-3 and described in Table 5-6.

Figure 5-3. TI OTP Bank 0 Package and Memory Size Information (F008 015Ch)

31 28	27 16
Reserved	PACKAGE
R	R

15 0
MEMORY_SIZE
R

LEGEND: R = Read only

Table 5-6. TI OTP Bank 0 Package and Memory Size Information Field Descriptions

Bit	Field	Description
31-28	Reserved	Reserved
27-16	PACKAGE	Count of pins in the package
15-0	MEMORY_SIZE	Flash memory size in Kbytes

5.4.2.3 LPO Trim and Max HCLK

The HF LPO trim solution, LF LPO trim solution and maximum HCLK frequency can be read from TI OTP location F008 01B4h as shown in Figure 5-4 and described in Table 5-7.

Figure 5-4. TI OTP Bank 0 LPO Trim and Max HCLK Information (F008 01B4h)

31 24	23 16
HFLPO_TRIM	LFLPO_TRIM
R	R

15 0
MAX_HCLK
R

LEGEND: R = Read only

Table 5-7. TI OTP Bank 0 LPO Trim and Max HCLK Information Field Descriptions

Bit	Field	Description
31-24	HFLPO_TRIM	HF LPO Trim Solution
23-16	LFLPO_TRIM	LF LPO Trim Solution
15-0	MAX_HCLK	Maximum HCLK Speed

5.4.2.4 Part Number Symbolization

Device part number symbolization information can be determined from the device-specific datasheet or can be computed by reading locations in the TI OTP bank 0 registers.

For example the device part number symbolization "TMS570LS3137CPGEQQ1" can be read from TI OTP bank 0 location F008 01E0h through F008 01FFh as shown in Figure 5-5.

Figure 5-5. TI OTP Bank 0 Symbolization Information (F008 01E0h-F008 01FFh)

0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F
0x54	0x4D	0x53	0x35	0x37	0x30	0x4C	0x53	0x33	0x31	0x33	0x37	0x43	0x50	0x47	0x45
R															

0x10	0x11	0x12	0x13	0x14	0x15	0x16	0x17	0x18	0x19	0x1A	0x1B	0x1C	0x1D	0x1E	0x1F
0x51	0x51	0x31	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00
R															

LEGEND: R = Read only

5.4.2.5 Deliberate ECC Errors for FMC ECC Checking

Deliberate single-bit and double-bit errors have been placed in the OTP for checking the FMC ECC functionality. Any portion of the 64 bits in TI OTP bank 0 location F008 03F0h through F008 03F7h as shown in Figure 5-6 will generate a single-bit error. Any portion of the 64 bits in TI OTP bank 0 location F008 03F8h through F008 03FFh as shown in Figure 5-6 will generate a double-bit error.

Figure 5-6. TI OTP Bank 0 Deliberate ECC Error Information (F008 03F0h-F008 03FFh)

0x00	0x04	0x08	0x0C
0x12345678	0x9ABCDEF1	0x12345678	0x9ABCDEF3
R	R	R	R

LEGEND: R = Read only, ECC is calculated for the value 0x123456789ABCDEF0

5.5 Power On, Power Off, and Reset Considerations

5.5.1 Error Checking at Power On

As the device is coming out of the device reset sequence, the Flash wrapper reads two configuration words from the TI OTP section of bank 0, the hardware configuration word at address 0xF0080140, and then the AJSM visible password at address 0xF0000000. During these reads ECC is enabled. Single-bit errors are corrected and generate an ESM group 1 channel 6 error event. The first failing address will be latched in the FCOR_ERR_ADD register along with the bit position in FCOR_ERR_POS register and the FEDACSTATUS register flags will be updated to indicate the type of error. Uncorrectable errors will generate an ESM group 3 channel 7 error event, the $\overline{\text{ERROR}}$ pin will be activated, the first failing address will be latched in the FUNC_ERR_ADD register and the FEDACSTATUS register flags will be updated to indicate the type of error.

5.5.2 Flash Integrity when Reset while Programming or Erasing

If a device is reset while programming, then the bits being programmed when reset is asserted are indeterminate; however, the other bits in the Flash are not disturbed. Likewise, If the device is reset while being erased, the sector or sectors being erased will have indeterminate bits; however, the other sectors in the same bank and the other banks will not be disturbed.

5.5.3 Flash Integrity at Power Off

If power is lost during a programming or erase operation, a power-on reset must be asserted before the core supply voltage drops below specification. The $\overline{\text{PORRST}}$ pin has a glitch filter which means that the $\overline{\text{PORRST}}$ pin must be asserted low $t_{f(nPORRST)}$ (2µs) before the core supply drops below Vcc_{MIN} (1.14V). If this requirement is met, then the bits being programmed when $\overline{\text{PORRST}}$ goes low are indeterminate; however, the other bits in the Flash are not disturbed. Likewise, if this requirement is met, and $\overline{\text{PORRST}}$ is asserted while erasing, the sector or sectors being erased will have indeterminate bits; however, the other sectors in the same bank and the other banks will not be disturbed.

5.6 Emulation and SIL3 Diagnostic Modes

5.6.1 System Emulation

During emulation when the SUSPEND signal is high, the data read from memory is still passed to SECDED for correction if ECC_ENABLE is active. If a correctable error is detected, then it is corrected but error event is not generated and error occurrence counter is not incremented if in profiling mode. If a double error is detected, then the raw data is returned without generating a double-error signal.

The SUSPEND signal can be disable by using the SUSP_IGNR bit in the Flash error detection and correction control register 1 (FEDACCTRL1). The SUSPEND signal should not be confused with the suspend_now operation for the FSM.

5.6.2 Diagnostic Mode

The Flash wrapper can be put in diagnostic mode to verify various logic. There are multiple diagnostic modes supported by the wrapper. A specific diagnostic mode is selected via the DIAG_MODE control bits in the diagnostic control register (FDIAGCTRL), as listed in Table 5-8.

The diagnostic mode is only enabled by a 4-bit key stored in the DIAG_EN_KEY bits in FDIAGCTRL register. Only DIAG_EN_KEY = 0101 enables any diagnostic mode and all diagnostic modes use the DIAG_TRIG bit in FDIAGCTRL register to initiate the action.

All tests run from any pipeline mode. Some of the diagnostic modes can corrupt the Flash data access, and generate errors as part of the test. Running in non-pipeline may minimize some of these conditions.

For all modes it is best to follow this sequence:

1. Write 0101 to the DIAG_EN_KEY bits and set the desired DIAG_MODE control bits. This blocks many UERR sources.
2. Set any data registers needed for this mode.
3. Write 1 to the DIAG_TRIG bit to initiate the action and allow UERRs to happen for one cycle.
4. Write 1010 to the DIAG_EN_KEY bits to disable the diagnostic modes.

When the CONF_TYPE is 5, the ECC logic is in the CPU and most diagnostic modes are removed. Some because the logic they test no longer exists and the rest because the path is validated via the ECC path to the CPU.

Table 5-8. DIAG_MODE Encoding

Mode	DIAG_MODE Bits			Description
0	0	0	0	Diagnostic mode is disabled. Same as DIAG_EN_KEY not equal to 5h.
1	0	0	1	ECC Data Correction test mode
2	0	1	0	ECC Syndrome Reporting test mode
3	0	1	1	ECC Malfunction test mode 1 (same data)
4	1	0	0	ECC Malfunction test mode 2 (inverted data)
5	1	0	1	Address Tag Register test mode
6	1	1	0	Reserved
7	1	1	1	ECC Data Correction Diagnostic test mode

5.6.2.1 ECC Data Correction Test Mode: DIAG_MODE = 1

This diagnostic mode can be enabled while ECC logic is also enabled for normal bank read. The Flash wrapper will arbitrate the usage of the ECC logic if a conflict occurs between a normal bank read and diagnostic checking.

When in diagnostic data correction mode, FEMU_xxxx registers contain the 64-bit EEPROM emulation data register, the 19-bit emulation address register and the 8-bit emulation check-bit register. These values are used to enter diagnostic data to exercise the SECDED logic. The user can apply a value with an error in any bit location. When the DIAG_TRIG is set, the SECDED calculation is done and the corrected values are saved back into the same FEMU_xxxx registers. The error position register is also updated to indicate the bit position in error. Either ERR_ONE_FLG or ERR_ZERO_FLG bit is set when a correctable error is detected. The D_COR_ERR bit will also be set in FEDACSTATUS register. For uncorrectable error, the error status bit ERR_PRF_FLG is set as well as the D_UNC_ERR bit in the same register. Status bits should be cleared by the user before applying a new diagnostic data.

It takes multiple CPU transactions to preload the registers with diagnostic values. During this time, the result of the diagnostic logic such as comparator can change. User should apply a trigger by setting DIAG_TRIG bit to 1 as a qualifier after all registers are loaded with intended values. The DIAG_TRIG serves to validate the diagnostic result. Only when DIAG_TRIG is high and a failing result in the diagnostic logic will update the corresponding status flag and the position register.

5.6.2.2 ECC Syndrome Reporting Test Mode: DIAG_MODE = 2

When in diagnostic syndrome reporting mode, the resulting syndrome calculated by SECDED is captured into the ECC check-bit register FEMU_ECC. The syndrome can be read by the user and compare with a known syndrome value. Diagnostic data in FEMU_DxSW and FEMU_ADDR is not corrected and the error position register is not updated. The FEDACSTATUS register error bits are not updated during this mode. See Figure 5-7.

For devices with ECC_IN_CPU (CONF_TYPE = 5), the resulting FEMU_ECC value represents the 32-bit byte swapped values. Here, bytes 7654_3210 are rearranged to 4567_0123. For instance, if the syndrome shows an error in data bit 33, it would really be an error in EMU_DMW bit 57. You can also XOR the data bit position with “011000”. (21h XOR 18h => 39h)

NOTE: The user should pre-load the registers with the test values with DIAG_TRIG = 0. After all test values are written, the DIAG_TRIG should then be set high to validate the diagnostic result.

5.6.2.3 ECC Malfunction Test Mode 1: DIAG_MODE = 3

There are three inputs to the malfunction detection logic: the resulting syndrome, the original uncorrected data, and the final corrected data. See Figure 5-7.

In normal function, the malfunction detection logic will detect an error if the syndrome is 0 and if the data before the correction and the data after the correction is not equal; or if the syndrome is not 0 and if the data before the correction and data after the correction is equal to each other. During diagnostic mode 3 or 4, user supplied values are sent to the malfunction logic. No functional checking is done by the ecc_malfunction logic while the mode is 3 or 4.

Diagnostic mode 3 is also known as “same data” mode. A diagnostic value can be stored in the ECC checkbit register. The value stored in the 64-bit Raw data register will be supplied to the two inputs of the malfunction comparator logic. If a non-zero value is stored in the Raw ECC checkbit register (FRAW_ECC), then the malfunction logic should detect it as an error and set the ECC_B2_MAL_ERR bit (ECC malfunction error). There is one ECC_B2_MAL_ERR bit for each SECDED block and they are called ECC1_MAL_ERR and ECC0_MAL_ERR and are selectable with the DIAG_ECC_SEL bit. The DIAG_TRIG is set to initiate this mode.

The FRAW_DATAx, FRAW_ECC, and FUNC_ERR_ADD will load on an ECC_MAL ERR but will not contain useful information during Diag mode 3.

5.6.2.4 ECC Malfunction Test Mode 2: DIAG_MODE = 4

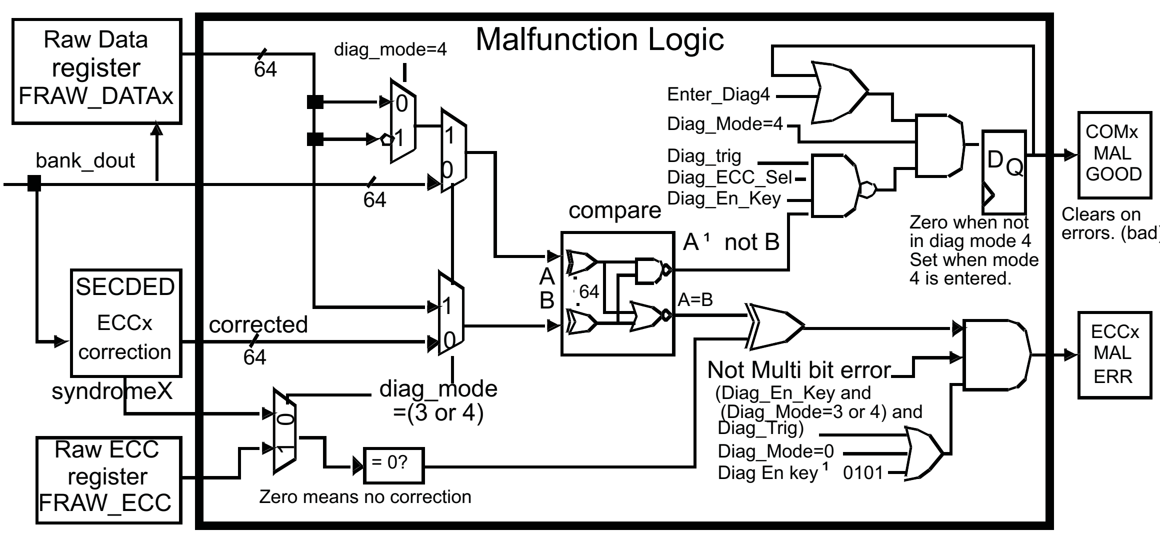
Diagnostic mode 4 is also known as “inverted data” mode. A diagnostic value can be stored in the ECC checkbit register. A value stored in the 64-bit Raw data register and its bit-wise inverted counterpart will be supplied to the two inputs of the malfunction comparator logic. If a zero value is stored in the Raw ECC checkbit register (FRAW_ECC), then the malfunction logic should detect it as an error and set the ECC_B2_MAL_ERR bit (diagnostic ECC malfunction). See Figure 5-7.

In this mode only, the EE_CMG bit (Compare Malfunction Good) is cleared if any one of the 64 XOR gates is malfunctioning. There is one EE_CMG bit for each SECDED block and they are called COM1_MAL_GOOD and COM0_MAL_GOOD. The EE_CMG bits go high when entering diagnostic mode 4 and go low and stay low if an error is detected. These bits are only valid in diagnostic mode 4 and outside of this mode the bits are 0.

Set the DIAG_ECC_SEL bits before entering mode 4 or enter mode 4 from a non-mode 4 and set the DIAG_ECC_SEL bits at the same time. Do not just change the DIAG_ECC_SEL bits or the corresponding MAL_GOOD will not get set because it did not enter mode 4 correctly.

The FRAW_DATAx, FRAW_ECC, and FUNC_ERR_ADD will load on an ECC_MAL ERR but will not contain useful information during Diag mode 4.

Figure 5-7. ECC Malfunction Test Logic



5.6.2.5 Address Tag Register Test Mode: DIAG_MODE = 5

NOTE: The test code for Diag mode 5 needs to be executed from RAM or when executed from Flash, the Flash Address Wait State (ASWSTEN) bit in the FRDCNTL register should be set to 1. This is due to conflicting accesses by the CPU and Flash wrapper during the test execution.

There are four sets of address tag registers. Each set consists of a primary and a duplicate address tag registers. Normally, these registers store the recently issued CPU addresses during pipeline mode. To detect errors in these registers, the primary and duplicate address tag registers are continuously compared to each other if the buffer is valid. If they are different, then an address tag register error event is generated.

These registers are memory-mapped. All primary address tag registers are memory-mapped to one address and, likewise, all duplicate tag registers are mapped to another single address. During diagnostic mode, each individual set can be selected by the DIAG_BUF_SEL (Diagnostic Buffer Select) bit in the FDIAGCTRL register. User-supplied values can be written into the selected set during a diagnostic mode. If different values are written into the primary and the duplicate address tag registers, then the ADD_TAG_ERR (Address Tag Error) flag in the FEDACSTATUS register will be set. This diagnostic mode uses the FRAW_DATAL register to supply the alternate address when DIAG_TRIG is set. The FUNC_ERR_ADD register will not contain useful information during Diag mode 5. It will also trigger the normal uncorrectable register freeze.

All address tags and buffer valid bits will be cleared to 0 when leaving Diag mode 5. Going to mode 5 and back out clears the pipeline buffers and is useful for other test modes also. No functional checking is done by the address tag logic while the mode is 5.

NOTE: The user should pre-load the registers with the test values with DIAG_TRIG = 0. After all test values are written, the DIAG_TRIG should then be set high to validate the diagnostic result.

5.6.2.6 ECC Data Correction Diagnostic Test Mode: DIAG_MODE = 7

Testing the error correction and ECC logic in the CPU involves corrupting the ECC value returned to the CPU. By inverting one or more bits of the ECC, the CPU will detect errors in a selected data or ECC bit, or in any possible value returned by the ECC.

To set an error for a particular bit use the syndrome, see Section 5.3.3. For example, if you want to corrupt data bit 62 then put the value 70h into the test register.

The method uses the DATA_INV_PAR value in the FPAR_OVR register to alter the ECC during a slave access cycle. The value in the DATA_INV_PAR register will be XORed with the current ECC to give a bad ECC value to the CPU. This only will occur when the DIAG_MODE is 7, the PAR_OVR_KEY is 5, the DIAG_EN_KEY in the FDIAGCTRL register is 5 and the access is a slave cycle.

This mode can set the FEDACSTATUS register status error bits B1_UNC_ERR or ERR_ZERO_FLG, but it will not set the D_UNC_ERR nor D_COR_ERR bits. Also, the logic to support the ECCx_MAL_ERR and COMx_MAL_GOOD bits is not implemented for the CPU path so these bits are not set.

The sequence to do this test is:

1. Make sure the true DMA module is off.
2. Put 5h in BUS_PAR_DIS and 5h in PAR_OVR_KEY fields (00005Axxh) of the FPAR_OVR register.
3. Put the desired value in DAT_INV_PAR field of the FPAR_OVR register.
4. Put 7h in DIAG_MODE and 5h in DIAG_EN_KEY fields of the FDIAGCTRL register.
5. Read desired address from the mirrored Flash location. Mirrored Flash starts at address 0x20000000.
6. Put 0 in DIAG_MODE or Ah in one of the key fields to turn off this test.
7. Check error registers (FCOR_ERR_ADD, FEDACSTATUS, and FUNC_ERR_ADD) for ECC errors.
8. Repeat as necessary to test out the ECC.
9. Put 0 in DIAG_MODE field of the FDIAGCTRL register and Ah in both of the key fields to completely disable this test at the end of the test.
10. Put 2h in PAR_OVR_KEY field (00005400h) of the FPAR_OVR register to clear DAT_INV_PAR field.

5.6.3 Diagnostic Mode Summary

The following tables give a summary of the input registers needed for each mode, the possible registers that can change, and the possible error bits in FEDACSTATUS register that may set.

Table 5-9. Bus 1 Diagnostic Mode Summary

DIAG MODE	Name	Inputs	Possible Outputs	Possible Error Bits Set	Notes
1	ECC Data Correction test mode				Not Applicable
2	ECC Syndrome Reporting test mode				Not Applicable
3	ECC Malfunction test mode 1				Not Applicable
4	ECC Malfunction test mode 2				Not Applicable
5	Address Tag Register test mode	FPRIM_ADD_TAG FDUP_ADD_TAG FRAW_DATAL	FUNC_ERR_ADD (1)	ADD_TAG_ERR	
6	Reserved				
7	ECC Data Correction Diagnostic test mode	DAT_INV_PAR	FUNC_ERR_ADD FCOR_ERR_ADD	B1_UNC_ERR ERR_ZERO_FLG	Slave access only

(1) Register output value will change, but will not contain useful information.

Table 5-10. Bus 2 and ECC Diagnostic Mode Summary

DIAG MODE	Name	Inputs	Possible Outputs	Possible Error Bits Set	Notes
1	ECC Data Correction test mode	FEUM_DMSW FEMU_DLSW FEMU_ECC FEMU_ADDR	FEMU_ECC FUNC_ERR_ADD FCOR_ERR_ADD FCOR_ERR_POS	D_UNC_ERR D_COR_ERR ERR_ONE_FLG ERR_ZERO_FLG ERR_PRF_FLG	
			FEMU_ECC EE_UNC_ERR_ADD EE_COR_ERR_ADD EE_COR_ERR_POS	EE_D_UNC_ERR EE_D_COR_ERR EE_ERR_ONE_FLG EE_ERR_ZERO_FLG EE_ERR_PRF_FLG	
2	ECC Syndrome Reporting test mode	FEMU_DMSW FEMU_DLSW FEMU_ECC FEMU_ADDR	FEMU_ECC	NA	
3	ECC Malfunction test mode 1	FRAW_DATAH FRAW_DATAL FRAW_ECC	FRAW_DATAH (1) FRAW_DATAL (1) FRAW_ECC (1) FUNC_ERR_ADD (1)	ECC_B2_MAL_ERR D_UNC_ERR	
			FRAW_DATAH (1) FRAW_DATAL (1) FRAW_ECC (1) EE_UNC_ERR_ADD (1)	EE_CME EE_D_UNC_ERR	

(1) Register output value will change, but will not contain useful information.

Table 5-10. Bus 2 and ECC Diagnostic Mode Summary (continued)

DIAG MODE	Name	Inputs	Possible Outputs	Possible Error Bits Set	Notes
4	ECC Malfunction test mode 2	FRAW_DATAH FRAW_DATAL FRAW_ECC	FRAW_DATAH [1] FRAW_DATAL [1] FRAW_ECC [1] FUNC_ERR_ADD [1]	COMB2_MAL_G ECC_B2_MAL_ERR D_UNC_ERR	
			FRAW_DATAH [1] FRAW_DATAL [1] FRAW_ECC [1] EE_UNC_ERR_ADD [1]	EE_CMG EE_CME EE_D_UNC_ERR	
5	Address Tag Register test mode				Not applicable
6	Reserved				Not applicable
7	ECC Data Correction Diagnostic test mode				Not applicable

Table 5-11. Port Signals Diagnostic Mode Summary

			Uncorrectable Error	Correctable Error	Address Bus Parity Error	FEE Uncorrectable Error	FEE Correctable Error
DIAG MODE	**Name**	**Error In**	**ESM Group 3 Channel 7**	**ESM Group 1 Channel 6**	**ESM Group 2 Channel 4**	**ESM Group 1 Channel 36**	**ESM Group 1 Channel 35**
1	ECC Data Correction test mode	Bus 2	Yes	Yes	No	No	No
		EEPROM	No	No	No	Yes	Yes
2	ECC Syndrome Reporting test mode	Bus 2	No	No	No	No	No
		EEPROM	No	No	No	No	No
3	ECC Malfunction test mode 1	Bus 2	Yes	No	No	No	No
		EEPROM	No	No	No	Yes	No
4	ECC Malfunction test mode 2	Bus 2	Yes	No	No	No	No
		EEPROM	No	No	No	Yes	No
5	Address Tag Register test mode	Bus 1	Yes	No	No	No	No
6	Reserved						
7	ECC Data Correction Diagnostic test mode	Bus 1	Yes	Yes	No	No	No

5.6.4 Read Margin

When the bits are programmed or erased, they are checked against a program_verify or erase_verify reference level that is far away from the normal read reference point. Over time, bit levels may drift toward the normal read point and if it is too much then a bit will read the wrong value. To counteract this, the bits can be read using different read_margin reference points to give an early detection of the problem. The bits can then be either re-programmed (most common) or the sector can be erased and reprogrammed.

5.7 Control Registers

This section details the Flash module registers, summarized in Table 5-12. A detailed description of each register and its bits is also provided.

The Flash module control registers can only be read and/or written by the CPU while in privileged mode. Each register begins on a word boundary. All registers are 32-bit, 16-bit and 8-bit accessible. The start address of the Flash module is FFF8 7000h.

Table 5-12. Flash Control Registers

Address	Acronym	Register Description	Section
FFF8 7000h	FRDCNTL	Flash Option Control Register	Section 5.7.1
FFF8 7008h	FEDACTRL1	Flash Error Detection and Correction Control Register 1	Section 5.7.2
FFF8 700Ch	FEDACTRL2	Flash Error Detection and Correction Control Register 2	Section 5.7.3
FFF8 7010h	FCOR_ERR_CNT	Flash Correctable Error Count Register	Section 5.7.4
FFF8 7014h	FCOR_ERR_ADD	Flash Correctable Error Address Register	Section 5.7.5
FFF8 7018h	FCOR_ERR_POS	Flash Correctable Error Position Register	Section 5.7.6
FFF8 701Ch	FEDACSTATUS	Flash Error Detection and Correction Status Register	Section 5.7.7
FFF8 7020h	FUNC_ERR_ADD	Flash Uncorrectable Error Address Register	Section 5.7.8
FFF8 7024h	FEDACSDIS	Flash Error Detection and Correction Sector Disable Register	Section 5.7.9
FFF8 7028h	FPRIM_ADD_TAG	Flash Primary Address Tag Register	Section 5.7.10
FFF8 702Ch	FDUP_ADD_TAG	Flash Duplicate Address Tag Register	Section 5.7.11
FFF8 7030h	FBPROT	Flash Bank Protection Register	Section 5.7.12
FFF8 7034h	FBSE	Flash Bank Sector Enable Register	Section 5.7.13
FFF8 7038h	FBBUSY	Flash Bank Busy Register	Section 5.7.14
FFF8 703Ch	FBAC	Flash Bank Access Control Register	Section 5.7.15
FFF8 7040h	FBFALLBACK	Flash Bank Fallback Power Register	Section 5.7.16
FFF8 7044h	FBPRDY	Flash Bank/Pump Ready Register	Section 5.7.17
FFF8 7048h	FPAC1	Flash Pump Access Control Register 1	Section 5.7.18
FFF8 704Ch	FPAC2	Flash Pump Access Control Register 2	Section 5.7.19
FFF8 7050h	FMAC	Flash Module Access Control Register	Section 5.7.20
FFF8 7054h	FMSTAT	Flash Module Status Register	Section 5.7.21
FFF8 7058h	FEMU_DMSW	EEPROM Emulation Data MSW Register	Section 5.7.22
FFF8 705Ch	FEMU_DLSW	EEPROM Emulation Data LSW Register	Section 5.7.23
FFF8 7060h	FEMU_ECC	EEPROM Emulation ECC Register	Section 5.7.24
FFF8 7068h	FEMU_ADDR	EEPROM Emulation Address Register	Section 5.7.25
FFF8 706Ch	FDIAGCTRL	Diagnostic Control Register	Section 5.7.26
FFF8 7070h	FRAW_DATAH	Uncorrected Raw Data High Register	Section 5.7.27
FFF8 7074h	FRAW_DATAL	Uncorrected Raw Data Low Register	Section 5.7.28
FFF8 7078h	FRAW_ECC	Uncorrected Raw ECC Register	Section 5.7.29
FFF8 707Ch	FPAR_OVR	Parity Override Register	Section 5.7.30
FFF8 70C0h	FEDACSDIS2	Flash Error Detection and Correction Sector Disable Register 2	Section 5.7.31
FFF8 7288h	FSM_WR_ENA	FSM Register Write Enable	Section 5.7.32
FFF8 72A4h	FSM_SECTOR	FSM Sector Register	Section 5.7.33
FFF8 72B8h	EEPROM_CONFIG	EEPROM Emulation Configuration Register	Section 5.7.34
FFF8 7308h	EE_CTRL1	EEPROM Emulation Error Detection and Correction Control Register 1	Section 5.7.35
FFF8 730Ch	EE_CTRL2	EEPROM Emulation Error Detection and Correction Control Register 2	Section 5.7.36
FFF8 7310h	EE_COR_ERR_CNT	EEPROM Emulation Correctable Error Count Register	Section 5.7.37
FFF8 7314h	EE_COR_ERR_ADD	EEPROM Emulation Correctable Error Address Register	Section 5.7.38
FFF8 7318h	EE_COR_ERR_POS	EEPROM Emulation Correctable Error Bit Position Register	Section 5.7.39
FFF8 731Ch	EE_STATUS	EEPROM Emulation Error Status Register	Section 5.7.40
FFF8 7320h	EE_UNC_ERR_ADD	EEPROM Emulation Uncorrectable Error Address Register	Section 5.7.41
FFF8 7400h	FCFG_BANK	Flash Bank Configuration Register	Section 5.7.42

5.7.1 Flash Option Control Register (FRDCNTL)

FRDCNTL supports pipeline mode. This register controls Flash timings for the main Flash banks. For the equivalent register that controls Flash timings for the EEPROM Emulation Flash bank (bank 7), see Section 5.7.34.

Figure 5-8. Flash Option Control Register (FRDCNTL) [offset = 00h]

31-16
Reserved
R-0

15-12	11-8
Reserved	RWAIT
R-0	R/WP-1

7-5	4	3-1	0
Reserved	ASWSTEN	Reserved	ENPIPE
R-0	R/WP-0	R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-13. Flash Option Control Register (FRDCNTL) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	RWAIT	0-Fh	Random/data Read Wait State
			The random read wait state bits indicate how many wait states are added to a Flash read access. In Pipeline mode there is always one wait state even when RWAIT is cleared to 0.
			Note: The required wait states for each HCLK frequency can be found in the device-specific data manual.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	ASWSTEN		Address Setup Wait State Enable
		0	Address Setup Wait State is disabled.
		1	Address Setup Wait State is enabled. Address is latched one cycle before decoding to determine pipeline hit or miss. Address Setup Wait State is only available in pipeline mode.
			Note: The required address wait state for each HCLK frequency can be found in the device-specific data manual.
3-1	Reserved	0	Reads return 0. Writes have no effect.
0	ENPIPE		Enable Pipeline Mode
		0	Pipeline mode is disabled.
		1	Pipeline mode is enabled.

5.7.2 Flash Error Detection and Correction Control Register 1 (FEDACCTRL1)

This register controls ECC event detection for the main Flash banks. For the equivalent register that controls ECC event detection for the EEPROM Emulation Flash bank (bank 7), see Section 5.7.35.

Figure 5-9. Flash Error Detection and Correction Control Register 1 (FEDACCTRL1) [offset = 08h]

31-25	24
Reserved	SUSP_IGNR
R-0	R/WP-0

23-20	19-16
Reserved	EDACMODE
R-0	R/WP-Ah

15-11	10	9	8
Reserved	EOFEN	EZFEN	EPEN
R-0	R/WP-0	R/WP-0	R/WP-0

7-4	3-0
Reserved	EDACEN
R-0	R/WP-5h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-14. Flash Error Detection and Correction Control Register 1 (FEDACCTRL1) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	SUSP_IGNR		Suspend Ignore
			In emulation mode, for example, viewing memory in the debugger's window, the CPU suspend signal is set. This bit determines whether the CPU suspend signal is ignored by the Flash module.
		0	CPU suspend signal blocks error bits setting and unfreezing.
			The Flash module blocks all errors from setting the error bits in emulation mode and blocks the unfreezing of the bits and registers by reading the FUNC_ERR_ADD register.
		1	CPU suspend has no effect on error bit setting and unfreezing.
			The Flash module ignores the CPU suspend signal and allows the error bits to set even in emulation mode. It also allows the Flash module to unfreeze the error bits and other registers by reading the FUNC_ERR_ADD register even in emulation mode.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	EDACMODE		Error Correction Mode for the main Flash banks. For EEPROM Emulation Flash bank (bank 7), see Section 5.7.35.
		5h	Single-bit errors during reads from OTP, ECC and the mirrored space (starting at 0x20000000) of banks 0 through 6, will be treated as uncorrectable errors by the Flash wrapper. The wrapper will assert an ESM group 3 error on channel 7 and the $\overline{\text{ERROR}}$ pin will be activated. No abort will be taken by the CPU.
		All Other Values	Single-bit errors during reads from OTP, ECC and the mirrored space (starting at 0x20000000) of banks 0 through 6, will be treated as correctable errors by the Flash wrapper. The wrapper will assert an ESM group 1 error on channel 6. The single-bit error will be corrected.
			Note: This mode does not affect reads from the main program Flash starting at address 0.
			Note: Reading ECC bits will generate an ECC error based on the contents of the 8 ECC bits and the 64 data bits they protect.
15-11	Reserved	0	Reads return 0. Writes have no effect.

Table 5-14. Flash Error Detection and Correction Control Register 1 (FEDACCTRL1) Field Descriptions (continued)

Bit	Field	Value	Description
10	EOFEN		Event on One's Fail Enable
		0	No ESM error event is generated on a single-bit error where a 1 reads as a 0 when reading from the OTP or ECC memory locations.
		1	An ESM error event is generated on a single-bit error where a 1 reads as a 0 when reading from the OTP or ECC memory locations.
			Note: When either the EOFEN or the EZFEN bit is set, an error event will be generated on ESM group 1 channel 6 when any correctable error is generated by reading the main memory.
9	EZFEN		Event on Zero's Fail Enable
		0	No ESM error event is generated on a single-bit error where a 0 reads as a 1 when reading from the OTP or ECC memory locations.
		1	An ESM error event is generated on a single-bit error where a 0 reads as a 1 when reading from the OTP or ECC memory locations.
			Note: When either the EOFEN or the EZFEN bit is set, an error event will be generated on ESM group 1 channel 6 when any correctable error is generated by reading the main memory.
8	EPEN		Error Profiling Enable.
		0	Error profiling is disabled.
		1	Error profiling is enabled.
			The correctable error event is generated (ESM group 1 channel 6) when the number of CPU accesses of correctable bit errors detected and corrected has reached the threshold value defined in the FEDACCTRL2 register.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	EDACEN		Error Detection and Correction Enable
		5h	CPU single and double error event signals are blocked.
			Note: It is recommended to enable ECC in the Flash wrapper by writing 1010 to these bits before enabling ECC in the CPU. If ECC is enabled in the CPU, but not in the wrapper, the CPU will still check and correct single-bit ECC errors, and generate aborts on uncorrectable errors for the main Flash. However, the generation of ESM events, the capture of failing addresses and the detections and correction of errors in the OTP will be prevented.
		All Other Values	Error Detection and Correction events are captured and sent to the ESM.

5.7.3 Flash Error Correction and Correction Control Register 2 (FEDACCTRL2)

This register applies to ECC event detection for the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.36.

Figure 5-10. Flash Error Correction and Correction Control Register 2 (FEDACCTRL2) [offset = 0Ch]

31	16
Reserved	
R-0	

15	0
SEC_THRESHOLD	
R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-15. Flash Error Correction Control and Correction Register 2 (FEDACCTRL2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	SEC_THRESHOLD		Single Error Correction Threshold When error profiling is enabled, this register contains the threshold value for the SEC (single error correction) occurrences before a correctable error event is generated (ESM group 1, channel 6). A threshold of zero disables the threshold so that it does not generate an event.

5.7.4 Flash Correctable Error Count Register (FCOR_ERR_CNT)

This register applies to the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.37.

Figure 5-11. Flash Correctable Error Count Register (FCOR_ERR_CNT) [offset = 10h]

31	16
Reserved	
R-0	

15	0
FERRCNT	
R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-16. Flash Correctable Error Count Register (FCOR_ERR_CNT) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	FERRCNT		Single Error Correction Count This register contains the number of SEC (single error correction) occurrences. Writing any value to this register resets the count value to 0. The counter resets to 0 when it increments to be equal to the single error correction threshold. This register only increments when profiling mode is enabled. This register is not affected by the EOFEN or EZEFEN error control bits in the FEDACCTRL1 register.

5.7.5 Flash Correctable Error Address Register (FCOR_ERR_ADD)

This register applies to the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.38.

The error address is captured during errors when either EOFEN or EZFEN enable bit is set. During error profiling mode when only EPEN is set, the error address is not captured if a correctable error is detected. This register is frozen while either the ERR_ZERO_FLG or the ERR_ONE_FLG bit is set in the FEDACSTATUS register.

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit (see Table 5-14), this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-12. Flash Correctable Error Address Register (FCOR_ERR_ADD) [offset = 14h]

31–16
COR_ERR_ADD
R-u

15–3	2–0
COR_ERR_ADD	B_OFF
R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-17. Flash Correctable Error Address Register (FCOR_ERR_ADD) Field Descriptions

Bit	Field	Value	Description
31-3	COR_ERR_ADD	0-1FFF FFFFh	Correctable Error Address COR_ERR_ADD records the CPU logical address of which a correctable error is detected by the ECC logic. This error address is frozen from begin updated until it is read by the CPU. Additional error are blocked until this register is read.
2-0	B_OFF	0-7h	Byte Offset Since ECC is checked on 64 bit data, when checking main memory or OTP, the address captured is aligned to a 64-bit boundary with address bits[2:0] equal to 0.When reading from the ECC bytes, these bits will indicate the failing address of the ECC location associated with the failure. When reading an ECC byte, the ECC is checked against the 64 data bits they protect.

5.7.6 Flash Correctable Error Position Register (FCOR_ERR_POS)

This register applies to the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.39.

Note: The bit error position is only detected during reads of the OTP, the mirrored Flash image or the ECC bytes. Single-bit errors corrected during reads of the main memory will only capture the failing address, but not the bit position. The bit position is captured during errors when either EOFEN or EZFEN enable bit is set. During error profiling mode when only EPEN is set, the bit position is not captured if a correctable error is detected. This register is frozen while either the ERR_ZERO_FLG or the ERR_ONE_FLG bit is set in the FEDACSTATUS register.

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit, this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-13. Flash Correctable Error Position Register (FCOR_ERR_POS) [offset = 18h]

31–16
Reserved
R-0

15–10	9	8	7–0
Reserved	BUS2	TYPE	ERR_POS
R-0	R-u	R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-18. Flash Correctable Error Position Register (FCOR_ERR_POS) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9	BUS2		Bus 2 Error
		0	The error was in the main Flash.
		1	The error was from an OTP read.
8	TYPE		ErrorType
		0	The error was one of the 64 data bits.
		1	The error was one of the 8 check bits.
7-0	ERR_POS		The bit address of the single-bit error.

5.7.7 Flash Error Detection and Correction Status Register (FEDACSTATUS)

This register applies to the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.40.

All these error status bits can be cleared by writing a 1 to the bit. Writing a 0 has no effect.

The correctable errors in bits 2:0, and FSM_DONE bit 24, must be cleared before the end of their error event service routine or else the error event will re-issue.

Figure 5-14. Flash Error Detection and Correction Status Register (FEDACSTATUS) [offset = 1Ch]

31-26	25	24
Reserved	Reserved	FSM_DONE
R-0	R-0	RCP-u

23-20	19	18	17	16
Reserved	COMB2_MAL_G	ECC_B2_MAL_ERR	B2_UNC_ERR	B2_COR_ERR
R-0	RCP-u	RCP-u	RCP-u	RCP-u

15-13	12	11	10	9	8
Reserved	D_UNC_ERR	ADD_TAG_ERR	ADD_PAR_ERR	Reserved	B1_UNC_ERR
R-0	RCP-u	RCP-u	RCP-u	R-0	RCP-u

7-4	3	2	1	0
Reserved	D_COR_ERR	ERR_ONE_FLG	ERR_ZERO_FLG	ERR_PRF_FLG
R-0	RCP-u	RCP-u	RCP-u	RCP-u

LEGEND: R = Read only; RCP = Read and Clear in Privilege Mode; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-19. Flash Error Detection and Correction Status Register (FEDACSTATUS) Field Descriptions

Bit	Field	Value	Description
31-26	Reserved	0	Reads return 0. Writes have no effect.
25	Reserved	0	Reserved
24	FSM_DONE		Flash State Machine Done This bit is set to 1 when the Flash state machine completes a program or erase operation. This bit will generate an interrupt on VIM channel 61 if the FSM_EVT_EN bit of the FSM_ST_MACHINE register is set. This bit must be cleared by writing a 1 to it in the interrupt routine to clear the interrupt request.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19	COMB2_MAL_G		Bus 2 Compare Malfunction Flag
		0	Compare Malfunction is detected on the Bus 2 SECDED in diagnostic mode 4 or is not in diagnostic mode.
		1	Compare Malfunction is not detected on the Bus 2 SECDED or entered diagnostic mode 4.
			This bit becomes 1 when entering diagnostic mode 4, with DIAG_ECC_SEL field set to 0 or 1, and will be cleared if diagnostic mode 4 triggers an error. This bit will reset to 0 and will be 0 outside of diagnostic mode 4. Writing a 1 will set this bit to 1 only in diagnostic mode 4; otherwise, writes have no effect.
18	ECC_B2_MAL_ ERR		Bus 2 ECC Malfunction Error Flag ECC_B2_MAL_ERR bit is set, if Bus 2 data is not corrected and a single-bit error is seen or data is corrected and a single-bit error is not seen.
		0	SECDED malfunction is not detected on Bus 2.
		1	SECDED malfunction is detected on Bus 2.

Table 5-19. Flash Error Detection and Correction Status Register (FEDACSTATUS) Field Descriptions (continued)

Bit	Field	Value	Description
17	B2_UNC_ERR		Bus 2 Uncorrectable Error Flag
		0	No bus 2 uncorrectable errors were detected.
		1	A bus 2 uncorrectable error was detected.
			Two or more bits in the data, or ECC field; or a single-bit error in the address field have been found in error. Address-bit errors are considered an uncorrectable error. The FUNC_ERR_ADD register should contain the Bus 2 error location. This error will generate an ESM group 3 channel 7 event.
16	B2_COR_ERR		Bus 2 Correctable Error Flag
		0	No bus 2 correctable error was detected.
		1	A bus 2 correctable error was detected.
			One bit in the data, or ECC field has been found in error. Either the ERR_ONE_FLAG or ERR_ZERO_FLAG should be set in this register along with this bit. The FCOR_ERR_ADD register should contain the error address, and the FCOR_ERR_POS register should contain the failing bit position. This error will generate an ESM group 1 channel 6 event.
15-13	Reserved	0	Reads return 0. Writes have no effect.
12	D_UNC_ERR		Diagnostic Uncorrectable Error Flag
			This bit sets when diagnostic mode 1 discovers a multi-bit error using the ECC. This means two or more bits in the data, address or ECC field have been found in error. The ECC is capable of correcting a single-bit error and this would show up in the D_COR_ERR bit. The ECC can always detect two bit errors. Three or more bit errors may escape detection with the ECC. This bit also may set during other uncorrectable errors and during the diagnostic mode like address tag errors and ECC malfunctions.
11	ADD_TAG_ERR		Address Tag Register Error Flag
		0	No address tag register error was detected.
		1	An address tag register error was detected.
			This bit is set if the primary address tag has a hit but the duplicate address tag does not match the primary address tag. This bit is functional only when pipeline mode is enabled. This error will create an ESM group 3 channel 7 event.
10	ADD_PAR_ERR		Address Parity Error Flag
		0	No address parity error was detected.
		1	A parity error was detected on the incoming address bus.
			The full 32 bit address will be stored in FUNC_ERR_ADD register. This error will create an ESM group 2 channel 4 event.
9	Reserved	0	Reads return 0. Writes have no effect.
8	B1_UNC_ERR		Bus 1 Uncorrectable Error Flag
		0	No bus 1 uncorrectable errors were detected.
		1	A bus 1 uncorrectable error was detected.
			Two or more bits in the data, or ECC field; or a single-bit error in the address field have been found in error. Address-bit errors are considered an uncorrectable error. The FUNC_ERR_ADD register will contain the Bus 1 error location. This error will generate an ESM group 3 channel 7 event..
7-4	Reserved	0	Reads return 0. Writes have no effect.
3	D_COR_ERR		Diagnostic Correctable Error Status Flag
			This bit sets when diagnostic mode 1 discovers a single-bit correctable error using the ECC. Multi-bit errors are flagged using the D_UNC_ERR bit. The uncorrectable error address must be unfrozen in order to set this bit.

Table 5-19. Flash Error Detection and Correction Status Register (FEDACSTATUS) Field Descriptions (continued)

Bit	Field	Value	Description
2	ERR_ONE_FLG		Error on One Fail Status Flag
		0	No correctable error where a 1 was read as a 0 on bus 2.
		1	A correctable error occurred on bus 2 where a 1 was read as a 0.
			This bit is set if the EOFEN (Error on One Fail Enable) bit is set then, and one bit in the data, or ECC field which should have been read as a 1 reads as a 0. During the read, the bit is corrected to a 1. The FCOR_ERR_ADD register will contain the bus 2 error address, and the FCOR_ERR_POS register will contain the failing bit position. This error will generate an ESM group 1 channel 6 event. When this bit is set, the B2_CORR_ERR bit will also be set. This error will generate an ESM group 1 channel 6 event.
1	ERR_ZERO_ FLG		Error on Zero Fail Status Flag
		0	No correctable errors on bus 1 nor any correctable errors on bus 2 where a 0 was read as a 1.
		1	A correctable error occurred on bus 1, or a correctable error occurred on bus 2 where a 0 was read as a 1.
			This bit is set if the EZFEN (Error on Zero Fail Enable) bit is set and a correctable error is detected on bus 2 where a 0 is read as a 1 and corrected to a 0, or if either the EZFEN or the EOFEN bits are set and any single-bit error is detected and corrected on bus 1. The FCOR_ERR_ADD register will contain the error address. If the error was on bus 2, then the B2_COR_ERR bit will also be set and the FCOR_ERR_POS register will contain the failing bit position. The FCOR_ERR_POS register will not indicate the failing bit position for a bus 1 error. This error will generate an ESM group 1 channel 6 event.
0	ERR_PRF_FLG		Error Profiling Status Flag
		0	Error profiling is not enabled, or the number of correctable errors has not reached the threshold programmed into the SEC_THRESHOLD register.
		1	Error profiling is enabled and the number of correctable errors has reached the threshold programmed into the SEC_THRESHOLD register.

5.7.8 Flash Uncorrectable Error Address Register (FUNC_ERR_ADD)

This register applies to ECC event detection for the main Flash banks. For the equivalent register that applies to the EEPROM Emulation Flash bank (bank 7), see Section 5.7.41.

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit, (see Table 5-14) this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-15. Flash Uncorrectable Error Address Register (FUNC_ERR_ADD) [offset = 20h]

31–16
UNC_ERR_ADD
R-u

15–3	2–0
UNC_ERR_ADD	B_OFF
R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-20. Flash Uncorrectable Error Address Register (FUNC_ERR_ADD) Field Descriptions

Bit	Field	Value	Description
31-3	UNC_ERR_ADD	0-1FFF FFFFh	Uncorrectable Error Address UNC_ERR_ADD records the CPU logical address of which an uncorrectable error is detected by the ECC logic in the CPU. The UNC_ERR_ADD also captures the error address when a address bus parity mismatch is detected. This error address is frozen from begin updated until it is read by the CPU. Additional error are blocked until this register is read. This register captures the full 32-bit incoming address when there is a bus parity error. It only captures address of 22:3 for multiple bit ECC errors. Address parity errors take priority over other errors that happen in the same cycle.
2-0	B_OFF	0-7h	Byte offset Since ECC is checked on 64 bit data, when checking main memory or OTP, the address captured is aligned to a 64-bit boundary with address bits[2:0] equal to 0.When reading from the ECC bytes, these bits will indicate the failing address of the ECC location associated with the failure. When reading an ECC byte, the ECC is checked against the 64 data bits they protect.

5.7.9 Flash Error Detection and Correction Sector Disable Register (FEDACSDIS)

This register is used to disable the SECDED function for one or two sectors from the EEPROM Emulation Flash (bank 7). An additional two sectors can have SECDED disabled by the use of the FEDACSDIS2 register (see Section 5.7.31).

Figure 5-16. Flash Error Detection and Correction Sector Disable Register (FEDACSDIS) [offset = 24h]

31-29	28	27-24	23-21	20	19-16
BankID1_Inverse	Rsvd	SectorID1_inverse	BankID1	Rsvd	SectorID1
R/WP-0	R-0	R/WP-0	R/WP-0	R-0	R/WP-0

15-13	12	11-8	7-5	4	3-0
BankID0_Inverse	Rsvd	SectorID0_inverse	BankID0	Rsvd	SectorID0
R/WP-0	R-0	R/WP-0	R/WP-0	R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-21. Flash Error Detection and Correction Sector Disable Register (FEDACSDIS) Field Descriptions

Bit	Field	Value	Description
31-29	BankID1_Inverse		The bank ID inverse bits are used with the bank ID bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		0	If BankID1 = 7h and BankID1_inverse = 0, then if a valid sector is selected by SectorID1 and SectorID1_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 1.
28	Reserved	0	Reads return 0. Writes have no effect.
27-24	SectorID1_inverse		The sector ID inverse bits are used with the sector ID bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 1.
23-21	BankID1		The bank ID bits are used with the bank ID inverse bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		7h	If BankID1 = 7h and BankID1_inverse = 0, then if a valid sector is selected by SectorID1 and SectorID1_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 1.
20	Reserved	0	Reads return 0. Writes have no effect.
19-16	SectorID1		The sector ID bits are used with the sector ID inverse bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 1.
15-13	BankID0_Inverse		The bank ID inverse bits are used with the bank ID bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		0	If BankID0 = 7h and BankID0_inverse = 0, then if a valid sector is selected by SectorID0 and SectorID0_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 0.
12	Reserved	0	Reads return 0. Writes have no effect.
11-8	SectorID0_inverse		The sector ID inverse bits are used with the sector ID bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 0.
7-5	BankID0		The bank ID bits are used with the bank ID inverse bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		7h	If BankID0 = 7h and BankID0_inverse = 0, then if a valid sector is selected by SectorID0 and SectorID0_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 0.
4	Reserved	0	Reads return 0. Writes have no effect.
3-0	SectorID0	0-Fh	The sector ID bits are used with the sector ID inverse bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 0.

5.7.10 Primary Address Tag Register (FPRIM_ADD_TAG)

This register is used to test the pipeline address tag registers (see Section 5.6.2.5).

Figure 5-17. Primary Address Tag Register (FPRIM_ADD_TAG) [offset = 28h]

31–16	
PRIM_ADD_TAG[31:16]	
R/WP-0	

15–4	3–0
PRIM_ADD_TAG[15:4]	0000
R/WP-0	R-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset;

Table 5-22. Primary Address Tag Register (FPRIM_ADD_TAG) Field Descriptions

Bit	Field	Value	Description
31-4	PRIM_ADD_TAG		Primary Address Tag Register The primary address tag register selected via DIAG_BUF_SEL bits in FDIAGCTRL register is memory mapped here. (see Section 5.7.26) This register can only be written in privileged mode when diagnostic mode is enabled with DIAG_EN_KEY = 5h and DIAG_MODE = 5h. This register will not update with new Flash data if DIAG_EN_KEY is not equal to 5h or DIAG_MODE is 0 or 7h. Valid reads can occur in any mode. The register will clear when an address tag error is found and when leaving DIAG_MODE 5.
3-0		0	Always 0000

5.7.11 Duplicate Address Tag Register (FDUP_ADD_TAG)

This register is used to test the pipeline address tag registers (see Section 5.6.2.5).

Figure 5-18. Duplicate Address Tag Register (FDUP_ADD_TAG) [offset = 2Ch]

31–16	
DUP_ADD_TAG[31:16]	
R/WP-0	

15–4	3–0
DUP_ADD_TAG[15:4]	0000
R/WP-0	R-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset;

Table 5-23. Duplicate Address Tag Register (FDUP_ADD_TAG) Field Descriptions

Bit	Field	Value	Description
31-4	DUP_ADD_TAG		Primary Address Tag Register The duplicate address tag register selected via DIAG_BUF_SEL bits in FDIAGCTRL register is memory mapped here. (see Section 5.7.26) This register can only be written in privileged mode when diagnostic mode is enabled with DIAG_EN_KEY = 5h and DIAG_MODE = 5h. This register will not update with new Flash data if DIAG_EN_KEY is not equal to 5h or DIAG_MODE is 0 or 7h. Valid reads can occur in any mode. The register will clear when an address tag error is found and when leaving DIAG_MODE 5.
3-0		0	Always 0000

5.7.12 Flash Bank Protection Register (FBPROT)

Figure 5-19. Flash Bank Protection Register (FBPROT) [offset = 30h]

31–16
Reserved
R-0

15–1	0
Reserved	PROTL1DIS
R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-24. Flash Bank Protection Register (FBPROT) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	PROTL1DIS		PROTL1DIS: Level 1 Protection Disabled
			Level 1 Protection Disable bit. Setting this bit disables protection from writing to the OTPPROTDIS bits as well as the Sector Enable registers FBSE for all banks. Clearing this bit enables protection and disables write access to the OTPPROTDIS register bits and FBSE register.
		0	Level 1 protection is enabled.
		1	Level 1 protection is disabled.

5.7.13 Flash Bank Sector Enable Register (FBSE)

FBSE provides one enable bit per sector for up to 16 sectors per bank. Each bank in the Flash module has one FBSE register. The bank is selected via the BANK[2:0] bits of the FMAC register (see Section 5.7.20). As only one bank at a time can be selected by FMAC, only the register for the bank selected appears at this address.

Figure 5-20. Flash Bank Sector Enable Register (FBSE) [offset = 34h]

31–16
Reserved
R-0

15–0
BSE
R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-25. Flash Bank Sector Enable Register (FBSE) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	BSE		Bank Sector Enable
			Each bit corresponds to a Flash sector in the bank specified by the FMAC register. Bit 0 corresponds to sector 0, bit 1 corresponds to sector 1, and so on. These bits can be set only when PROTL1DIS = 1 and in privilege mode.
		0	The corresponding numbered sector is disabled for program or erase access.
		1	The corresponding numbered sector is enabled for program or erase access.

5.7.14 Flash Bank Busy Register (FBBUSY)

Figure 5-21. Flash Bank Busy Register (FBBUSY) [offset = 38h]

31		16
Reserved		
R-0		

15	8	7	0
		BUSY[7:0]	
R/WP-0		R-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-26. Flash Bank Busy Register (FBBUSY) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	BUSY[7:0]		Bank Busy
			Each bit corresponds to a Flash bank.
		0	The corresponding bank is not busy.
		1	The corresponding bank is busy with a state machine or bus 2 operation, or the bank is not implemented.

5.7.15 Flash Bank Access Control Register (FBAC)

Figure 5-22. Flash Bank Access Control Register (FBAC) [offset = 3Ch]

31–24	23–16
Reserved	OTPPROTDIS
R-0	R/WP-0

15–8	7–0
BAGP	VREADST
R/WP-0	R/WP-Fh

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-27. Flash Bank Access Control Register (FBAC) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	OTPPROTDIS		OTP Sector Protection Disable
			Each bit corresponds to a Flash bank. This bit can be set only when PROTL1DIS = 1 and in privilege mode.
		0	Programming of the OTP sector is disabled.
		1	Programming of the OTP sector is enabled.
15-8	BAGP	0-FFh	Bank Active Grace Period
			These bits contain the starting count value for the BAGP down counter. Any access to a given bank causes its BAGP counter to reload the BAGP value for that bank. After the last access to this Flash bank, the down counter delays from 0 to 255 prescaled HCLK clock cycles before putting the bank into one of the fallback power modes as determined by the FBFALLBACK register. This value must be greater than 1 when the fallback mode is not ACTIVE.
			Note: The prescaled clock used for the BAGP down counter is a clock divided by 16 from HCLK.
7-0	VREADST	0-FFh	VREAD Setup
			VREAD is generated by the Flash pump and used for Flash read operation. The bank power up sequencing starts VREADST HCLK cycles after VREAD power supply becomes stable.
			Note: There is not a programmable Bank Sleep counter and Standby counter register. The number of clock cycles to transition from sleep to standby and standby to active is hardcoded in the Flash wrapper design.

5.7.16 Flash Bank Fallback Power Register (FBFALLBACK)

Figure 5-23. Flash Bank Fallback Power Register (FBFALLBACK) [offset = 40h]

31–16
Reserved
R-0505h

15–14	13–4	3–2	1–0
BANKPWR7	Reserved	BANKPWR1	BANKPWR0
R/WP-3h	R/WP-3FFh	R/WP-3h	R/WP-3h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-28. Flash Bank Fallback Power Register (FBFALLBACK) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0505h	Do not write to these bits.
15-14	BANKPWR7		Bank 7 Fallback Power Mode
		0	Bank sleep mode
		1h	Bank standby mode
		2h	Reserved
		3h	Bank active mode
13-4	Reserved	3FFh	Do not write to these bits.
3-2	BANKPWR1		Bank 1 Fallback Power Mode
		0	Bank sleep mode
		1h	Bank standby mode
		2h	Reserved
		3h	Bank active mode
1-0	BANKPWR0		Bank 0 Fallback Power Mode
		0	Bank sleep mode
		1h	Bank standby mode
		2h	Reserved
		3h	Bank active mode

5.7.17 Flash Bank/Pump Ready Register (FBPRDY)

Figure 5-24. Flash Bank/Pump Ready Register (FBPRDY) [offset = 44h]

31–24	23–16
Reserved	BANKBUSY[7:0]
R-0	R-1 (Unimplemented banks) or R-0 (Implemented banks)

15	14–8	7–0
PUMPRDY	Reserved	BANKRDY[7:0]
R-1	R-0	R-1

LEGEND: R = Read only; -*n* = value after reset

Table 5-29. Flash Pump Access Control Register 1 (FPAC1) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	BANKBUSY[7:0]		Bank busy bits (one bit for each bank)
		0	The bank is not busy.
		1	The bank is busy, not ready or this bank is not implemented.
			Note: A bank is considered busy if it is being accessed by the TCM, Bus 2 or the Flash state machine.
15	PUMPRDY		Flash pump ready flag
		0	Pump is not ready (Code must be executing from somewhere other than internal Flash).
		1	Pump is ready for Flash accesses.
14-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	BANKRDY[7:0]		Bank ready bits (one bit for each bank)
		0	Flash bank is in the sleep or standby state.
		1	Flash bank is in the active state, or the bank is not implemented.

5.7.18 Flash Pump Access Control Register 1 (FPAC1)

Figure 5-25. Flash Pump Access Control Register 1 (FPAC1) [offset = 48h]

31–27	26–16
Reserved	PSLEEP
R-0	R/WP-64h

15–1	0
Reserved	PUMPPWR
R-0	R/WP-1

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-30. Flash Pump Access Control Register 1 (FPAC1) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
26-16	PSLEEP		Pump Sleep These bits contain the starting count value for the charge pump sleep down counter. While the charge pump is in sleep mode, the power mode management logic holds the charge pump sleep counter at this value. When the charge pump exits sleep power mode, the down counter delays from 0 to PSLEEP pump sleep down clock cycles before putting the charge pump into active power mode. **Note:** Pump sleep down counter clock is a divide by 2 input of HCLK. That is, there are 2*HCLK cycles for every PSLEEP counter cycle.
15-1	Reserved	0	Reads return 0. Writes have no effect.
0	PUMPPWR		Flash Charge Pump Fallback Power Mode
		0	Sleep (all pump circuits disabled).
		1	Active (all pump circuits active).

5.7.19 Flash Pump Access Control Register 2 (FPAC2)

Figure 5-26. Flash Pump Access Control Register 2 (FPAC2) [offset = 4Ch]

31–16
Reserved
R-0

15–0
PAGP
R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-31. Flash Pump Access Control Register 2 (FPAC2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	PAGP		Pump Active Grace Period This register contains the starting count value for the PAGP mode down counter. Any access to Flash memory causes the counter to reload with the PAGP value. After the last access to Flash memory, the down counter delays from 0 to 65535 prescaled HCLK clock cycles before entering one of the charge pump fallback power modes as determined by PUMPPWR in the FPAC1 register. **Note:** The PAGP down counter is clocked by the same prescaled clock as the BAGP down counter which is a divide by 16 of HCLK.

5.7.20 Flash Module Access Control Register (FMAC)

Figure 5-27. Flash Module Access Control Register (FMAC) [offset = 50h]

31–16
Reserved
R-0

15–3	2–0
Reserved	BANK
R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-32. Flash Module Access Control Register (FMAC) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
2-0	BANK		Bank Enable. These bits select which bank is enabled for operations such as local register access, OTP sector access, and program/erase commands. These bits select only one bank at a time from up to eight banks depending on the specific device being used. For example, a 000 selects bank 0; 011 selects Bank 3. **Note:** BANK[2:0] can identify up to 8 Flash banks. If BANK[2:0] is selected for an unimplemented bank then the BANK[2:0] will set itself to the number of an implemented bank. To determine if a bank is implemented, write the bank number to BANK[2:0] and read back the value to see if what was written can be read back.

5.7.21 Flash Module Status Register (FMSTAT)

Figure 5-28. Flash Module Status Register (FMSTAT) [offset = 54h]

31–24
Reserved
R-0

23–16
Reserved
R-0

15	14	13	12	11	10	9	8
Reserved	ILA	Reserved	PGV	Reserved	EV	Reserved	BUSY
R-0	R-0	R-0	R-0	R-0	R-0	R-0	R-0

7	6	5	4	3	2	1	0
ERS	PGM	INVDAT	CSTAT	VOLTSTAT	ESUSP	PSUSP	SLOCK
R-0	R-0	R-0	R-0	R-0	R-0	R-0	R-0

LEGEND: R = Read only; -*n* = value after reset

Table 5-33. Flash Module Status Register (FMSTAT) Field Descriptions

Bit	Field	Value	Description
31-15	Reserved	0	Reads return 0. Writes have no effect.
14	ILA		Illegal Address When set, indicates that an illegal address is detected. Five conditions can set the illegal address flag. 1. Writing to a hole (unimplemented logical address space) within a Flash bank. 2. Writing to an address location to an unimplemented Flash space. 3. Input address for write is decoded to select a different bank from the bank ID register. 4. The address range does not match the type of FSM command. For example, the erase_sector command must match the address regions. 5. TI-OTP address selected but CMD_EN in FSM_ST_MACHINE is not set.
13	Reserved	0	Reads return 0. Writes have no effect.
12	PGV		Program Verify When set, indicates that a word is not successfully programmed after the maximum allowed number of program pulses are given for program operation.
11	Reserved	0	Reads return 0. Writes have no effect.
10	EV		Erase Verify When set, indicates that a sector is not successfully erased after the maximum allowed number of erase pulses are given for erase operation. During Erase verify command, this flag is set immediately if a bit is found to be 0.
9	Reserved	0	Reads return 0. Writes have no effect.
8	BUSY		Busy When set, this bit indicates that a program, erase, or suspend operation is being processed.
7	ERS		Erase Active When set, this bit indicates that the Flash module is actively performing an erase operation. This bit is set when erasing starts and is cleared when erasing is complete. It is also cleared when the erase is suspended and set when the erase resumes.
6	PGM		Program Active When set, this bit indicates that the Flash module is currently performing a program operation. This bit is set when programming starts and is cleared when programming is complete. It is also cleared when programming is suspended and set when programming is resumes.
5	INVDAT		Invalid Data When set, this bit indicates that the user attempted to program a 1 where a 0 was already present. This bit is cleared by the Clear Status command.

Table 5-33. Flash Module Status Register (FMSTAT) Field Descriptions (continued)

Bit	Field	Value	Description
4	CSTAT		Command Status Once the FSM starts any failure will set this bit. When set, this bit informs the host that the program, erase, or validate sector command failed and the command was stopped. This bit is cleared by the Clear Status command. For some errors, this will be the only indication of an FSM error because the cause does not fall within the other error bit types.
3	VOLTSTAT		Core Voltage Status When set, this bit indicates that the core voltage generator of the pump power supply dipped below the lower limit allowable during a program or erase operation. This bit is cleared by the Clear Status command.
2	ESUSP		Erase Suspended When set, this bit indicates that the Flash module has received and processed an erase suspend operation. This bit remains set until the erase resume command has been issued or until the Clear_More command is run.
1	PSUSP		Program Suspended When set, this bit indicates that the Flash module has received and processed a program suspend operation. This bit remains set until the program resume command has been issued or until the Clear_More command is run.
0	SLOCK		Sector Lock Status When set, this bit indicates that the operation was halted because the target sector was locked for erasing and programming either by the sector protect bit or by OTP write protection disable bits. (Bits BSE in FBSE register or OTPPROTDIS in register FBAC). This bit is cleared by the Clear Status command. No SLOCK FSM error will occur if all sectors in a bank erase operation are set to 1. All the sectors will be checked but no SLOCK will be set if no operation occurs due to the SECT_ERASED bits being set to all 1s. A SLOCK error will occur if attempting to do a sector erase with either BSE is cleared or SECT_ERASED is set.

5.7.22 EEPROM Emulation Data MSW Register (FEMU_DMSW)

The Flash module controller includes hardware support computing the check bits for ECC, based on the data and address being programmed into the EEPROM emulation array. To utilize this capability, the address to be programmed is written to the FEMU_ADDR register and the data to be programmed is written to the FEMU_DMSW and FEMU_DLSW registers. The write to FEMU_DLSW triggers an ECC calculation and the resulting check bits are available in the FEMU_ECC register. The value from FEMU_ECC can then be used to program the check bits into the EEPROM emulation array for the particular data word over which they were calculated.

Figure 5-29. EEPROM Emulation Data MSW Register (FEMU_DMSW) [offset = 58h]

31 ... 0
EMU_DMSW[63:32]
R/WP-0

LEGEND: R/W = Read/Write; WP = Write in Privilege mode; -*n* = value after reset

Table 5-34. EEPROM Emulation Data MSW Register (FEMU_DMSW) Field Descriptions

Bit	Field	Description
31-0	EMU_DMSW	EEPROM Emulation Most-Significant Data Word. The most-significant data word of the 64-bits of data for which the ECC check bits are to be calculated should be programmed into this register. This register is also used in diagnostic modes 1 and 2 where it supplies the upper data for checking the SECDED hardware.

5.7.23 EEPROM Emulation Data LSW Register (FEMU_DLSW)

The Flash module controller includes hardware support computing the check bits for ECC, based on the data and address being programmed into the EEPROM emulation array. To utilize this capability, the address to be programmed is written to the FEMU_ADDR register and the data to be programmed is written to the FEMU_DMSW and FEMU_DLSW registers. The write to FEMU_DLSW triggers an ECC calculation and the resulting check bits are available in the FEMU_ECC register. The value from FEMU_ECC can then be used to program the check bits into the EEPROM emulation array for the particular data word over which they were calculated.

Figure 5-30. EEPROM Emulation Data LSW Register (FEMU_DLSW) [offset = 5Ch]

31 ... 0
EMU_DLSW[31:0]
R/WP-0

LEGEND: R/W = Read/Write; WP = Write in Privilege mode; -*n* = value after reset

Table 5-35. EEPROM Emulation Data LSW Register (FEMU_DLSW) Field Descriptions

Bit	Field	Description
31-0	EMU_DLSW	EEPROM Emulation Least-Significant Data Word. The least-significant data word of the 64-bits of data for which the ECC check bits are to be calculated should be programmed into this register. This register is also used in diagnostic modes 1 and 2 where it supplies the lower data for checking the SECDED hardware.

5.7.24 EEPROM Emulation ECC Register (FEMU_ECC)

The Flash module controller includes hardware support computing the check bits for ECC, based on the data and address being programmed into the EEPROM emulation array. To utilize this capability, the address to be programmed is written to the FEMU_ADDR register and the data to be programmed is written to the FEMU_DMSW and FEMU_DLSW registers. The write to FEMU_DLSW triggers an ECC calculation and the resulting check bits are available in the FEMU_ECC register. The value from FEMU_ECC can then be used to program the check bits into the EEPROM emulation array for the particular data word over which they were calculated.

Figure 5-31. EEPROM Emulation ECC Register (FEMU_ECC) [offset = 60h]

31-16
Reserved
R-0

15-8	7-0
Reserved	EMU_ECC[7:0]
R-0	R/WP-3h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege mode; -*n* = value after reset

Table 5-36. EEPROM Emulation ECC Register (FEMU_ECC) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reserved
7-0	EMU_ECC[7:0]		EEPROM Emulation ECC Check Bit Value. This register contains the ECC check bits calculated by the FMC controller based on the address written to FEMU_ADDR and the 64-bits of data written to FEMU_DMSW and FEMU_DLSW. This register is also used in the diagnostic modes 1 and 2. In these modes, this register supplies the ECC data for checking the SECDED. In mode 1, this register is filled with the desired ECC and after the DIAG_TRIG is set this register is set to the ECC value returned from the SECDED selected by DIAG_ECC_SEL. DIAG_EN_KEY and DIAG_TRIG must be set to fill this register in the diagnostic modes. Writes to FEMU_DxSW will not affect this register in diagnostic modes 1 or 2. In mode 2, this register is filled with the desired ECC and after the DIAG_TRIG is set this register is set to the syndrome value returned from the SECDED selected by DIAG_ECC_SEL. This register is only available when the module is configured to use the ECC logic; otherwise, it is a reserved register.

5.7.25 EEPROM Emulation Address Register (FEMU_ADDR)

The Flash module controller includes hardware support computing the check bits for ECC, based on the data and address being programmed into the EEPROM emulation array. To utilize this capability, the address to be programmed is written to the FEMU_ADDR register and the data to be programmed is written to the FEMU_DMSW and FEMU_DLSW registers. The write to FEMU_DLSW triggers an ECC calculation and the resulting check bits are available in the FEMU_ECC register. The value from FEMU_ECC can then be used to program the check bits into the EEPROM emulation array for the particular data word over which they were calculated.

Figure 5-32. EEPROM Emulation Address Register (FEMU_ADDR) [offset = 68h]

31-22	21-16
Reserved	EMU_ADDR[21:16]
R-0	R/WP-0

15-3	2-0
EMU_ADDR[15:3]	Reserved
R/WP-0	R-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege mode; -*n* = value after reset

Table 5-37. EEPROM Emulation Address Register (FEMU_ADDR) Field Descriptions

Bit	Field	Value	Description
31-22	Reserved	0	Reserved. Writes have no effect. It is not necessary to mask these upper ten bits when writing the address of bank 7 locations (0xF002xxxx); however, these bits are not used in calculating the ECC value and will read back as 0.
21-3	EMU_ADDR	0-7 FFFFh	EEPROM Emulation Address. The address of the 64-bit data word over which ECC is to be calculated is written to this field. Note that only bits 21:3 are actually written and used for the calculation. The other bits (31:22 and 2:0) are ignored, but do not need to be masked off before being written to this register. This register is also used in diagnostic modes 1 and 2 where it supplies the address bits for checking the SECDED hardware.
2-0	Reserved	0	Reserved. Writes have no effect. The lower three bits of the CPU address are not used in the ECC calculation to align the data on a 64-bit boundary. These bits will read back as 0.

5.7.26 Diagnostic Control Register (FDIAGCTRL)

Set the DIAG_MODE and the DIAG_EN_KEY first before setting up the other registers to block the other registers from causing a false error. The final write should set the DIAG_TRIG register to activate the test. Running out of ram will prevent problems with the diagnostic test corrupting the Flash access in some of the modes.

Figure 5-33. Diagnostic Control Register (FDIAGCTRL) [offset = 6Ch]

31-25	24	23-20	19-16
Reserved	DIAG_TRIG	Reserved	DIAG_EN_KEY
R-0	R/WP-0	R-0	R/WP-Ah

15	14-12	11-10	9-8	7-3	2-0
Rsvd	DIAG_ECC_SEL	Reserved	DIAG_BUF_SEL	Reserved	DIAG_MODE
R-0	R/WP-0	R-0	R/WP-0	R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege mode; -*n* = value after reset

Table 5-38. Diagnostic Control Register (FDIAGCTRL) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reserved
24	DIAG_TRIG		Diagnostic Trigger Diagnostic trigger is the final qualifier for the diagnostic result. After setting all the other diagnostic register values, the DIAG_TRIG is set to 1. This will activate the diagnostic logic for one access and then automatically clear the DIAG_TRIG value. DIAG_EN_KEY and DIAG_MODE must be set at least one cycle before setting DIAG_TRIG. This bit always reads as 0.
23-20	Reserved	0	Reserved
19-16	DIAG_EN_KEY		Diagnostic Enable Key
		5h	Diagnostic mode is enabled.
		All other values	Diagnostic mode is disabled.
15	Reserved	0	Reserved
14-12	DIAG_ECC_SEL		Diagnostic SECDED Select
		0	Select SECDED0 for diagnostic testing.
		1h	Select SECDED1 for diagnostic testing.
		2h	Select SECDED2 for diagnostic testing (256 bit wide words only).
		3h	Select SECDED3 for diagnostic testing (256 bit wide words only).
		4h	Select BUS2 SECDED for diagnostic testing.
		5h	Select FEE SECDED for diagnostic testing (Same ECC logic as BUS2 but sets FEE registers).
		6h-7h	Reserved
11-10	Reserved	0	Reserved

Table 5-38. Diagnostic Control Register (FDIAGCTRL) Field Descriptions (continued)

<table>
<tr><th>Bit</th><th>Field</th><th>Value</th><th>Description</th></tr>
<tr><td rowspan="2">9-8</td><td rowspan="2">DIAG_BUF_ SEL</td><td></td><td>Diagnostic Buffer Select</td></tr>
<tr><td>0</td><td>The DIAG_BUF_SEL selects the Instruction or Data buffer to read or write when accessing the FPRIM_ADD_TAG and FDUP_ADD_TAG registers. The address tags consists of matching primary and duplicate address tag registers. All the primary address tag registers are memory mapped to a common address (see Section 5.7.10) and are selected by DIAG_BUF_SEL. The same occurs for the duplicate address. (see Section 5.7.11). Bit 0 selects a data buffer if high and an instruction buffer if low. Bits 1 indicate the buffer number.
DIAG_BUF_SEL ENCODING:
<table>
<tr><th>Buffer Number Bits 1</th><th>Inst=0 Data=1 Bit0</th><th>Buffer</th></tr>
<tr><td>0</td><td>0</td><td>Instruction Buffer 0</td></tr>
<tr><td>0</td><td>1</td><td>Data Buffer 0</td></tr>
<tr><td>1</td><td>0</td><td>Instruction Buffer 1</td></tr>
<tr><td>1</td><td>1</td><td>Data Buffer 1</td></tr>
</table></td></tr>
<tr><td>7-4</td><td>Reserved</td><td>0</td><td>Reserved</td></tr>
<tr><td rowspan="9">2-0</td><td rowspan="9">DIAG_MODE</td><td></td><td>Diagnostic Mode</td></tr>
<tr><td>0</td><td>Diagnostic mode is disabled. This is the same as DIAG_EN_KEY is not equal to 5h.</td></tr>
<tr><td>1h</td><td>Diagnostic ECC Data Correction test mode (see Section 5.6.2.1).</td></tr>
<tr><td>2h</td><td>Diagnostic ECC Syndrome Reporting test mode (see Section 5.6.2.2).</td></tr>
<tr><td>3h</td><td>ECC Malfunction test mode 1 (same data) (see Section 5.6.2.3).</td></tr>
<tr><td>4h</td><td>ECC Malfunction test mode 2 (inverted data) (see Section 5.6.2.4).</td></tr>
<tr><td>5h</td><td>Address Tag Register test mode (see Section 5.6.2.5).</td></tr>
<tr><td>6h</td><td>Reserved</td></tr>
<tr><td>7h</td><td>ECC Data Correction Diagnostic test mode (see Section 5.6.2.6).</td></tr>
</table>

5.7.27 Uncorrected Raw Data High Register (FRAW_DATAH)

Figure 5-34. Uncorrected Raw Data High Register (FRAW_DATAH) [offset = 70h]

31	0
RAW_DATA[63:32]	
R/WP-u	

LEGEND: R/W = Read/Write; WP = Write in Privilege mode; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-39. Uncorrected Raw Data High Register (FRAW_DATAH) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA [63:32]	Uncorrected Raw Data This register contains the upper 32 bits of the 64-bit raw data used in diagnostic testing of the ECC logic. **NOTE:** Raw Data and Raw ECC registers can be loaded with diagnostic values only in diagnostic modes 1 through 6 with DIAG_EN_KEY=0101. These modes must be set for at least one clock cycle before writing to any FRAW* register.

5.7.28 Uncorrected Raw Data Low Register (FRAW_DATAL)

Figure 5-35. Uncorrected Raw Data Low Register (FRAW_DATAL) [offset = 74h]

31	0
RAW_DATA[31:0]	
R/WP-u	

LEGEND: R/W = Read/Write; WP = Write in Privilege mode; -*n* = value after reset -u = unchanged value on internal reset, cleared on power up

Table 5-40. Uncorrected Raw Data Low Register (FRAW_DATAL) Field Descriptions

Bit	Field	Description
31-0	RAW_DATA [31:0]	Uncorrected Raw Data. This register contains the lower 32 bits of the 64-bit raw data used in diagnostic testing of the ECC logic. **NOTE:** Raw Data and Raw ECC registers can be loaded with diagnostic values only in diagnostic modes 1 through 6 with DIAG_EN_KEY=0101. These modes must be set for at least one clock cycle before writing to any FRAW* register.

5.7.29 Uncorrected Raw ECC Register (FRAW_ECC)

Figure 5-36. Uncorrected Raw ECC Register (FRAW_ECC) [offset = 78h]

31–16
Reserved
R-0

15–9	8	7–0
Reserved	PIPE BUF	RAW_ECC[7:0]
R-0	RC-0	R/WD-u

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege mode; C = Clear by writing a 1; -*n* = value after reset -u = unchanged value on internal reset, cleared on power up

Table 5-41. Uncorrected Raw ECC Register (FRAW_ECC) Field Descriptions

Bit	Field	Value	Description
31-9	Reserved	0	Reserved
8	PIPE BUF		Error came from pipeline buffer hit When this bit is a 1, latest error came from a pipeline buffer hit and the FRAW_DATH, FRAW_DATL, and RAW_ECC fields will not contain information that matches the error address nor error status bits. This bit is cleared when the RAW_ECC field is updated with new valid information or by writing a 1 to this bit.
7-0	RAW_ECC[7:0]		Uncorrected Raw ECC This register contains the ECC data used in diagnostic testing of the ECC logic. **NOTE:** Raw Data and Raw ECC registers can loaded with diagnostic values only in a used diagnostic mode with DIAG_EN_KEY=0101. This mode must be set for at least one clock cycle before writing to any FRAW* register.

5.7.30 Parity Override Register (FPAR_OVR)

Figure 5-37. Parity Override Register (FPAR_OVR) [offset = 7Ch]

31–17	16
Reserved	BNK_INV_PAR
R-0	R/WP-0

15–12	11–9	8	7–0
BUS_PAR_DIS	PAR_OVR_KEY	ADD_INV_PAR	DAT_INV_PAR
R/WP-5h	R/WP-2h	R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-42. Parity Override Register (FPAR_OVR) Field Descriptions

Bit	Field	Value	Description
31-17	Reserved	0	Reserved
16	BNK_INV_PAR		Buffer Invert Parity When this value is 1 and PAR_OVR_KEY is 101 then the current system parity signal SYS_ODD_PARITY is inverted when doing bank parity calculations. This will generate parity errors and cause interrupt signals to be generated. When 0, the SYS_ODD_PARITY value is used. This bit is only implemented for parity configurations and is reserved for ECC devices.
15-12	BUS_PAR_DIS		Disable Bus Parity When this value is 1010, the address bus parity error and buffer parity error are disabled and no checking is done and no events are generated. Any other value will enable the parity checking on the Address bus and read data bus. The read data parity is never disabled from this module.
11-9	PAR_OVR_KEY		When this value is 101, the selected ADD_INV_PAR and DAT_INV_PAR fields will become active. Any other value will cause the module to use the global system parity bit in the system register DEVCR1.
8	ADD_INV_PAR		Address Odd Parity This bit is active only when PAR_OVR_KEY = 101. When ADD_INV_PAR is 1, the incoming address bus will invert the system signal SYS_ODD_PARITY for parity calculations. This will cause parity errors and generate interrupt error signals. When 0, it will use the SYS_ODD_PARITY value. This bit is set to the SYS_ODD_PAR signal value on reset.
7-0	DAT_INV_PAR		Data Odd Parity This byte is active only when PAR_OVR_KEY = 101. When a DATA_INV_PAR bit is 1, the output read data will invert the system signal SYS_ODD_PARITY for parity calculations. This will cause parity errors and generate interrupt signals. When 0, it will use the SYS_ODD_PARITY value. This byte can support up to a 64 bit data bus but when the device has a 32 bit bus, bits 7:4 are reserved. Bit 0 affects read bus bits 7:0, Bit 1 affects read bus bits 15:8 and so on. Each active bit of this field is set to the SYS_ODD_PAR signal value on reset. The DAT_INV_PAR is used in the parity for the pipeline buffer logic and for the read data bus to the CPU. When the ECC logic is in the CPU (CONF_TYPE = 5) and SIL3 is active, this field becomes the ECC corrupting value for SIL3 diagnostic mode 7. (Starting with version 1.0.0.0) In diagnostic mode 7 the FPAR_OVR should be set to 00005Axxh to allow writes to the DAT_INV_PAR field. This field should be written before entering diagnostic mode 7.

5.7.31 Flash Error Detection and Correction Sector Disable Register (FEDACSDIS2)

This register is used to disable the SECDED function for one or two sectors from the EEPROM Emulation Flash (bank 7). An additional two sectors can have SECDED disabled by the use of the FEDACSDIS register. see Section 5.7.9.

Figure 5-38. Flash Error Detection and Correction Sector Disable Register (FEDACSDIS2) [offset = C0h]

31-29	28	27-24	23-21	20	19-16
BankID3_Inverse	Rsvd	SectorID3_inverse	BankID3	Rsvd	SectorID3
R/WP-0	R-0	R/WP-0	R/WP-0	R-0	R/WP-0

15-13	12	11-8	7-5	4	3-0
BankID2_Inverse	Rsvd	SectorID2_inverse	BankID2	Rsvd	SectorID2
R/WP-0	R-0	R/WP-0	R/WP-0	R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-43. Flash Error Detection and Correction Sector Disable Register (FEDACSDIS2) Field Descriptions

Bit	Field	Value	Description
31-29	BankID3_Inverse		The bank ID inverse bits are used with the bank ID bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		0	If BankID3 = 7h and BankID3_inverse = 0, then if a valid sector is selected by SectorID3 and SectorID3_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 3.
28	Reserved	0	Reads return 0. Writes have no effect.
27-24	SectorID3_inverse		The sector ID inverse bits are used with the sector ID bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 3.
23-21	BankID3		The bank ID bits are used with the bank ID inverse bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		7h	If BankID3 = 7h and BankID3_inverse = 0, then if a valid sector is selected by SectorID3 and SectorID3_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 3.
20	Reserved	0	Reads return 0. Writes have no effect.
19-16	SectorID3		The sector ID bits are used with the sector ID inverse bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 3.
15-13	BankID2_Inverse		The bank ID inverse bits are used with the bank ID bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		0	If BankID2 = 7h and BankID2_inverse = 0, then if a valid sector is selected by SectorID2 and SectorID2_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 2.
12	Reserved	0	Reads return 0. Writes have no effect.
11-8	SectorID2_inverse		The sector ID inverse bits are used with the sector ID bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 2.
7-5	BankID2		The bank ID bits are used with the bank ID inverse bits to select the bank for which a sector is disabled. The only bank that supports sector disable is bank 7.
		7h	If BankID2 = 7h and BankID2_inverse = 0, then if a valid sector is selected by SectorID2 and SectorID2_inverse that sector will have ECC checking disabled.
		other	No sector is disabled by disable ID 2.
4	Reserved	0	Reads return 0. Writes have no effect.
3-0	SectorID2	0-Fh	The sector ID bits are used with the sector ID inverse bits to determine which sector is disabled. If the sector ID bits are not pointing to a valid sector (0-3) or the sector ID inverse bits are not an inverse of the sector ID bits, then no sector is disabled by disable ID 2.

5.7.32 FSM Register Write Enable (FSM_WR_ENA)

Figure 5-39. FSM Register Write Enable (FSM_WR_ENA) [offset = 288h]

31–16
Reserved
R-0

15–3	2–0
Reserved	WR_ENA
R-0	R/WP-2h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-44. FSM Register Write Enable (FSM_WR_ENA) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
2-0	WR_ENA		Flash State Machine Write Enable
		5h	This register must contain 101 in order to write to any other register in the range FFF8 7200h to FFF8 72FFh. This is the first register to be written when setting up the FSM.
		All other values	For all other values, the FSM registers cannot be written.

5.7.33 FSM Sector Register (FSM_SECTOR)

This is a banked register. A separate register is implemented for each bank, but they all occupy the same address. The correct bank must be selected in the FMAC register before reading or writing this register. See Section 5.7.20.

Figure 5-40. FSM Sector Register (FSM_SECTOR) [offset = 2A4h]

31–16
SECT_ERASED[16:0]
R/WP-0

15–0
Reserved
R-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-45. FSM Sector Register (FSM_SECTOR) Field Descriptions

Bit	Field	Value	Description
31-16	SECT_ERASED[16:0]		Sectors Erased
			There is one bit for each sector. Bit 16 corresponds to sector 0, bit 17 corresponds to sector 1, and so on. After bank erase, the bit corresponding to each sector which is erased will be changed from 0 to 1.
		0	During bank erase, each sector will be erased.
		1	Each sector will not be erased.
15-0	Reserved	0	These bits are used by the state machine during bank erase. Do not write to these bits.

5.7.34 EEPROM Emulation Configuration Register (EEPROM_CONFIG)

Figure 5-41. EEPROM Emulation Configuration Register (EEPROM_CONFIG) [offset = 2B8h]

31–20	19–16
Reserved	EWAIT
R-0	R/WP-1h

15–9	8	7–0
Reserved	AUTOSUSP_EN	AUTOSTART_GRACE
R-0	R/WP-0	R/WP-2h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-46. EPROM Emulation Configuration Register (EEPROM_CONFIG) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	EWAIT		EEPROM Wait State Counter
			This register will replace the RWAIT count in the EEPROM register. The same formulas that apply to RWAIT will apply to EWAIT in the EEPROM bank.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	AUTOSUSP_EN		Auto Suspend Enable
		0	Auto Suspend is disabled.
		1	Auto Suspend is enabled.
			The auto-suspend will begin when the CPU or Bus 2 attempts to access a bank with an active and suspendable FSM operation. If this happens the FSM will automatically be issued a suspend command and exit from the FSM. It will then do the access. After the access, the FMC will wait for a time determined by the Autostart_grace field before issuing the FSM resume command.
7-0	AUTOSTART_GRACE		Auto-suspend Startup Grace Period
		1	The value in this register determines how many cycles the FMC will wait after the last CPU or Bus 2 access before issuing the FSM resume command.
		0	The FMC will wait 16 HCLK periods for each count in the AUTOSTART_GRACE field. A value of 2 will wait for 32 periods after the last access. Each access will reset the counter to the AUTOSTART_GRACE value × 16.

5.7.35 EEPROM Emulation Error Detection and Correction Control Register 1 (EE_CTRL1)

This register controls ECC event detection for the EEPROM Emulation Flash bank (bank 7). For the equivalent register that controls ECC event detection for the main Flash banks, see Section 5.7.2.

Figure 5-42. EEPROM Emulation Error Detection and Correction Control Register 1 (EE_CTRL1) [offset = 308h]

31–24
Reserved
R-05h

23–20	19–16
Reserved	EE_EDACMODE
R-0	R/WP-Ah

15–11	10	9	8
Reserved	EE_EOFEN	EE_EZFEN	EE_EPEN
R-0	R/WP-0	R/WP-0	R/WP-0

7–6	5	4	3–0
Reserved	EE_ALL1_OK	EE_ALL0_OK	EE_EDACEN
R-0	R/WP-0	R/WP-0	R/WP-5h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-47. EEPROM Emulation Error Detection and Correction Control Register 1 (EE_CTRL1) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	050h	Reads return 050h. Writes have no effect.
19-16	EE_ EDACMODE		Error Correction Mode for the EEPROM Emulation Flash bank (bank 7). For the main Flash banks, see Section 5.7.2.
		5h	Detection only mode.
			Note: In detection only mode single-bit errors will not be corrected, but will be treated as uncorrectable errors. The single-bit error flags and profiling mode are disabled. Detection only mode has the advantage that a triple bit error will be detected and not mistaken for a single-bit error and mis-corrected.
		All Other Values	Single-bit errors are corrected and multi-bit or address errors are detected.
			Note: It is recommended to leave the EE_EDACMODE field as 1010 to guard against soft errors from flipping EE_EDACMODE to a detection only.
15-11	Reserved	0	Reads return 0. Writes have no effect.
10	EE_EOFEN		EEPROM Emulation Event on a correctable One's Fail Enable bit
		0	No ESM event will be generated on a single-bit error when a 1 reads as a 0 and is corrected.
		1	An ESM group 1 channel 35 event will be generated on a single-bit error when a 1 reads as a 0 and is corrected.
9	EE_EZFEN		EEPROM Emulation Event on a correctable Zero's Fail Enable bit
		0	No ESM event will be generated on a single-bit error when a 0 reads as a 1 and is corrected.
		1	An ESM group 1 channel 35 event will be generated on a single-bit error when a 0 reads as a 1 and is corrected.
8	EE_EPEN		EEPROM Emulation Error Profiling Enable
		0	Error profiling is disabled.
		1	Error profiling is enabled.
			An ESM group 1 channel 35 event will be generated when number of correctable bit errors detected and corrected has reached the threshold value defined in the EE_CTRL2 register.
7-6	Reserved	0	Reads return 0. Writes have no effect.

Table 5-47. EEPROM Emulation Error Detection and Correction Control Register 1 (EE_CTRL1) Field Descriptions (continued)

Bit	Field	Value	Description
5	EE_ALL1_OK		EEPROM Emulation All One Condition Valid
		0	One condition valid is disabled.
			Reading of an erased location (64 data bits and the corresponding 8 ECC bits are all 1s) will generate ECC errors. The error counter for profiling will increment if all 1s are detected.
		1	One condition valid is enabled.
			Reading of an erased location (64 data bits and the corresponding 8 ECC bits are all 1s) will NOT generate ECC errors. The error counter for profiling will NOT increment if all 1s are detected.
4	EE_ALL0_OK		EEPROM Emulation All Zero Condition Valid
		0	Zero condition valid is disabled.
			Reading of all 0s (64 data bits and the corresponding 8 ECC bits are all 0s) will generate ECC errors. The error counter for profiling will increment if all 0s are detected.
		1	Zero condition valid is enabled.
			Reading of all 0s (64 data bits and the corresponding 8 ECC bits are all 0s) will NOT generate ECC errors. The error counter for profiling will NOT increment if all 0s are detected.
3-0	EE_EDACEN		EEPROM Emulation Error Detection and Correction Enable
		5h	Error Detection and Correction is disabled.
		All Other Values	Error Detection and Correction is enabled.
			Note: It is recommended to leave the EE_EDACEN field as 1010 to guard against soft errors from flipping the EE_EDACEN to a disabled state.

5.7.36 EEPROM Emulation Error Correction and Correction Control Register 2 (EE_CTRL2)

Figure 5-43. EEPROM Emulation Error Correction and Correction Control Register 2 (EE_CTRL2) [offset = 30Ch]

31	16
Reserved	
R-0	

15	0
EE_SEC_THRESHOLD	
R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-48. EEPROM Emulation Error Correction Control Register 2 (EE_CTRL2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	EE_SEC_THRESHOLD	0-FFFFh	EEPROM Emulation Single Error Correction Threshold This register contains the threshold value for the SEC (single error correction) occurrences before a single interrupt request is generated. A threshold of zero disables the threshold so that it never triggers the profile interrupt.

5.7.37 EEPROM Emulation Correctable Error Count Register (EE_COR_ERR_CNT)

Figure 5-44. EEPROM Emulation Error Correctable Error Count Register (EE_COR_ERR_CNT) [offset = 310h]

31	16
Reserved	
R-0	

15	0
EE_ERRCNT	
R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privilege Mode; -*n* = value after reset

Table 5-49. EEPROM Emulation Correctable Error Count Register (EE_COR_ERR_CNT) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	EE_ERRCNT	0-FFFFh	Single Error Correction Count This register contains the number of SEC (single error correction) occurrences. Writing any value to this register resets the count value to 0. The counter resets to 0 when it increments to be equal to the single error correction threshold. This register only increments when profiling mode is enabled. This register is not affected by the EE_ZERO_EN or EE_ONE_EN error control bits in the EE_CTRL1 register.

5.7.38 EEPROM Emulation Correctable Error Address Register (EE_COR_ERR_ADD)

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit, (seeTable 5-14) this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-45. EEPROM Emulation Correctable Error Address Register (EE_COR_ERR_ADD) [offset = 314h]

31–16
COR_ERR_ADD
R-u

15–3	2–0
COR_ERR_ADD	B_OFF
R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-50. EEPROM Emulation Correctable Error Address Register (EE_COR_ERR_ADD) Field Descriptions

Bit	Field	Value	Description
31-3	COR_ERR_ADD	0-1FFF FFFFh	Correctable Error Address COR_ERR_ADD records the CPU logical address of which a correctable error is detected by the ECC logic. This error address is frozen from begin updated until it is read by the CPU. Additional error are blocked until this register is read.
2-0	B_OFF	0-7h	Byte offset Since ECC is checked on 64 bit data, when checking main memory or OTP, the address captured is aligned to a 64-bit boundary with address bits[2:0] equal to 0. When reading from the ECC bytes, these bits will indicate the failing address of the ECC location associated with the failure. When reading an ECC byte, the ECC is checked against the 64 data bits they protect.

5.7.39 EEPROM Emulation Correctable Error Position Register (EE_COR_ERR_POS)

The bit position is captured during errors when either EE_EOFEN or EE_EZFEN enable bit is set. During error profiling mode when only EE_EPEN is set, the bit position is not captured if a correctable error is detected. This register is frozen while either the EE_ERR_ZERO_FLG or the EE_ERR_ONE_FLG bit is set in the EE_EDACSTATUS register.

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit, this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-46. EEPROM Emulation Correctable Error Position Register (EE_COR_ERR_POS) [offset = 318h]

31-16
Reserved
R-0

15-8	8	7-0
Reserved	TYPE	EE_ERR_POS
R-0	R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-51. EEPROM Emulation Correctable Error Position Register (EE_COR_ERR_POS) Field Descriptions

Bit	Field	Value	Description
31-9	Reserved	0	Reads return 0. Writes have no effect.
8	TYPE		Error Type
		0	The error was one of the 64 data bits.
		1	The error was one of the 8 check bits.
7-0	EE_ERR_POS	0-FFh	The bit address of the single-bit error.

5.7.40 EEPROM Emulation Error Status Register (EE_STATUS)

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit in FEDACCTRL1 register, (see Table 5-14) this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

All these error status bits can be deactivated by writing a 1 to the bit. Writing a 0 has no effect.

Bits 0 to 3 show correctable errors while bits 4 to 12 show uncorrectable errors. When the uncorrectable errors are triggered, the current address is stored in the EE_UNC_ERR_ADD register.

These error bits are not set while the FMC is in the suspend mode but they can be cleared in suspend by writing 1s to the bits. By setting the SUSP_IGNR bit to 1, these error bits can be set in suspend mode (see Table 5-14).

Figure 5-47. EEPROM Emulation Error Status Register (EE_STATUS) [offset = 31Ch]

31-24
Reserved
R-0

23-16
Reserved
R-0

15-13	12	11-9	8
Reserved	EE_D_UNC_ERR	Reserved	EE_UNC_ERR
R-0	RCP-u	R-0	RCP-u

7	6	5	4	3	2	1	0
Reserved	EE_CMG	Reserved	EE_CME	EE_D_COR_ERR	EE_ERR_ONE_FLG	EE_ERR_ZERO_FLG	EE_ERR_PRF_FLG
R-0	R-0	R-0	R-0	RCP-u	RCP-u	RCP-u	RCP-u

LEGEND: R = Read only; RCP = Read and Clear in Privilege Mode; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-52. EEPROM Emulation Error Status Register (EE_STATUS) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12	EE_D_UNC_ERR		Diagnostic Mode Uncorrectable Error Status Flag
		0	No uncorrectable error was detected in diagnostic mode 1.
		1	An uncorrectable error was detected in diagnostic mode 1. This means two or more bits in the data or ECC field have been found in error, or one or more bits in the address have been found in error.
11-9	Reserved	0	Reads return 0. Writes have no effect.
8	EE_UNC_ERR		EEPROM Emulation Uncorrectable Error Flag
		0	No uncorrectable errors were detected in bank 7.
		1	An uncorrectable error was detected in bank 7.
7	Reserved	0	Reads return 0. Writes have no effect.
6	EE_CMG		EEPROM Emulation Compare Malfunction Good
		0	Compare malfunction was detected on the Bus2 SECDED logic.
		1	Compare malfunction was not detected on the Bus2 SECDED logic.
5	Reserved	0	Reads return 0. Writes have no effect.
4	EE_CME		EEPROM Emulation Compare Malfunction Error
		0	Compare malfunction was not detected on the Bus2 SECDED logic.
		1	Compare malfunction was detected on the Bus2 SECDED logic.

Table 5-52. EEPROM Emulation Error Status Register (EE_STATUS) Field Descriptions (continued)

Bit	Field	Value	Description
3	EE_D_COR_ERR		Diagnostic Correctable Error Flag
		0	No correctable error was detected.
		1	A correctable error was detected.
2	EE_ERR_ONE_FLG		Error on One Fail Error Flag
		0	No correctable error was detected.
		1	A correctable error was detected.
1	EE_ERR_ZERO_FLG		Error on Zero Fail Error Flag
		0	No correctable error was detected.
		1	A correctable error was detected.
0	EE_ERR_PRF_FLG		Error Profiling Error Flag
		0	No correctable error was detected.
		1	A correctable error was detected.

5.7.41 EEPROM Emulation Uncorrectable Error Address Register (EE_UNC_ERR_ADD)

During emulation mode, this address is frozen even when read. By setting the SUSP_IGNR bit, (see Table 5-14) this register can be unfrozen in emulation mode.

This register is not changed with the reset signal and contains unknown data at power-up.

Figure 5-48. EEPROM Emulation Uncorrectable Error Address Register (EE_UNC_ERR_ADD) [offset = 320h]

31–16
UNC_ERR_ADD
R-u

15–3	2–0
UNC_ERR_ADD	B_OFF
R-u	R-u

LEGEND: R = Read only; -*n* = value after reset; -u = unchanged value on internal reset, cleared on power up

Table 5-53. EEPROM Emulation Uncorrectable Error Address Register (EE_UNC_ERR_ADD) Field Descriptions

Bit	Field	Value	Description
31-3	UNC_ERR_ADD	0-1FFF FFFFh	Uncorrectable Error Address UNC_ERR_ADD records the CPU logical address of which an uncorrectable error is detected by the ECC logic. This error address is frozen from begin updated until it is read by the CPU. Additional error are blocked until this register is read.
2-0	B_OFF	0-7h	Byte offset Since ECC is checked on 64-bit data, when checking main memory or OTP, the address captured is aligned to a 64-bit boundary with address bits[2:0] equal to 0.When reading from the ECC bytes, these bits will indicate the failing address of the ECC location associated with the failure. When reading an ECC byte, the ECC is checked against the 64 data bits they protect.

5.7.42 Flash Bank Configuration Register (FCFG_BANK)

Figure 5-49. Flash Bank Configuration Register (FCFG_BANK) [offset = 400h]

31–20	19–16
EE_BANK_WIDTH	Reserved
R-90h	R-1

15–4	3–0
MAIN_BANK_WIDTH	Reserved
R-90h	R-2h

LEGEND: R = Read only; -*n* = value after reset

Table 5-54. Flash Bank Configuration Register (FCFG_BANK) Field Descriptions

Bit	Field	Value	Description
31-20	EE_BANK_WIDTH	90h	Bank 7 width (144-bits wide) This read-only value indicates the maximum number of bits that can be programmed in the bank in one operation. The 144 bits includes 128 data bits and 16 ECC bits.
19-16	Reserved	1	Writes have no effect.
15-4	MAIN_BANK_WIDTH	90h	Width of main Flash banks (144-bits wide) This read-only value indicates the maximum number of bits that can be programmed in the bank in one operation. The 144 bits includes 128 data bits and 16 ECC bits.
3-0	Reserved	2h	Writes have no effect.